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RESEARCH

Ultra-low communication overhead federated learning strategy for distributed wearable ECG networks

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Abstract

Federated learning (FL) over wearable electrocardiogram (ECG) networks enables privacy-preserving cardiac monitoring at scale, yet standard FL protocols are incompatible with the severe bandwidth and energy constraints of heterogeneous LoRa IoT deployments. This paper proposes ECG-FedCS, a hardware-aware FL framework that dynamically modulates gradient sparsification intensity based on real-time ECG signal quality, QRS complex morphology, and heart-rate variability indices extracted on-device. Nodes operating in high-quality clinical environments transmit a larger gradient fraction, preserving feature diversity for accurate multi-class cardiac classification; bandwidth-constrained LoRa nodes receive protocol-aware maximum compression via a hardware-aware factor, jointly optimizing diagnostic accuracy and energy efficiency. ECG-FedCS is evaluated on a federated simulation of ten heterogeneous nodes spanning BLE and LoRa wireless links, using patient records from the PhysioNet CinC 2020 dataset across nine cardiac conditions. The proposed framework achieves a classification accuracy of 91.24%, converging faster than all baselines while achieving an 80.8 × per-parameter compression ratio. Per-round energy consumption is substantially reduced, extending LoRa node communication-related battery lifetime by 2.4 × under baseline (idealized) channel conditions; a sensitivity analysis under realistic packet-delivery degradation (Section VI-H) shows this relative multiplier is preserved, and increases further for the most distance-constrained rural nodes as channel conditions degrade. ECG-FedCS simultaneously outperforms all four baselines — FedAvg, FedProx, QSGD, and TopK — in accuracy, convergence speed, and compression ratio, establishing the first FL framework to achieve this Pareto improvement in a multi-protocol wearable cardiac IoT setting. These results demonstrate that ECG-signal guidance enables privacy-preserving, communication-efficient federated cardiac monitoring on resource-constrained remote nodes, advancing arrhythmia detection in low-connectivity clinical environments.

Keywords Federated Learning · Wearable ECG · Gradient Sparsification · LoRa · Differential Privacy

1 Introduction

Cardiovascular disease remains the leading global cause of mortality, accounting for an estimated 19.8 million deaths annually (World Health Organization 2024). Continuous ambulatory electrocardiogram (ECG) monitoring via wearable IoT devices offers a transformative pathway for early arrhythmia detection, enabling timely clinical intervention in populations without routine access to hospital-grade diagnostic infrastructure. The deployment of such systems, however, is constrained by two fundamental tensions: the clinical imperative to share patient-generated data sufficiently broadly to train accurate deep learning models, and the legal and ethical imperative to preserve patient privacy under frameworks such as the Health Insurance Portability and Accountability Act (HIPAA) and the General Data Protection Regulation (GDPR) (World Health Organization 2024).

Federated learning (FL) has emerged as the dominant paradigm for reconciling these competing demands (McMahan et al. 2017). In an FL framework, each participating node trains a local model replica on its private patient data and transmits only gradient updates — rather than raw ECG signals — to a central aggregation server. The server aggregates these updates via a weighted averaging rule and broadcasts the refined global model, thereby enabling collaborative learning without direct data sharing (Yang et al.

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2019). Despite this architectural advantage, standard FL protocols such as FedAvg (McMahan et al. 2017) and FedProx (Li et al. 2020) transmit full 32-bit floating-point gradient vectors at every communication round. For wearable ECG nodes operating over bandwidth-constrained wireless links — particularly LoRa networks with raw data rates as low as 0.73 kbps and transmission energy costs exceeding 34,000 nJ/bit — this constitutes an untenable communication overhead.

Existing gradient compression methods address this limitation from two complementary directions: quantization-based approaches such as QSGD (Alistarh et al. 2017) reduce per-gradient bit depth to 8 bits, achieving a 4× compression ratio; and sparsification-based approaches such as Deep Gradient Compression (DGC) (Lin et al. 2018) transmit only the top 0.1% of gradient magnitudes, achieving up to 600× compression. Critically, however, neither approach is aware of the clinical signal content being processed. When a wearable node observes an acute ST-elevation myocardial infarction (STEMI) pattern, the associated gradients carry disproportionately high diagnostic information, and aggressive compression may suppress precisely the updates most critical for downstream model generalization. Conversely, during periods of normal sinus rhythm — which constitute approximately 35.3% of all recordings in the present study — substantially higher compression is clinically safe, as the diagnostic information density is correspondingly lower.

A comprehensive review of 28 biosignal- and EHR-based FL studies published in 2025 (Ryu et al. 2025) confirms that none of the surveyed methods simultaneously addresses gradient compression and LoRa energy constraints in wearable cardiac IoT networks. Similarly, recent adaptive sparsification frameworks for general FL tasks (Moradi Kalarde et al. 2025; Hönig et al. 2022) have extended compression ratios or reduced communication overhead substantially, yet remain signal-agnostic and have not been evaluated in ECG-specific or energy-constrained wearable deployments — the precise gap that ECG-FedCS addresses.

This paper introduces ECG-FedCS, the first FL gradient compression framework that dynamically modulates sparsification intensity based on real-time ECG signal features. The central intuition is that the optimal compression ratio for a given gradient update is not a fixed engineering constant but a function of the clinical content of the underlying ECG signal. Nodes operating in high-quality clinical environments — characterized by elevated SQI scores reflecting reliable ECG acquisition — are assigned a higher sparsity ratio, transmitting a larger gradient fraction that preserves feature diversity for diagnostically complex cases; nodes in bandwidth-constrained LoRa environments receive additional protocol-aware compression via the hardware-aware γ factor, maximizing energy efficiency while maintaining convergence quality. This content-aware strategy is operationalized through a lightweight on-device feature extraction module that computes QRS duration, HRV indices (SDNN and RMSSD), peak amplitude, and signal SNR in real time, producing a scalar sparsity ratio that governs gradient masking for each communication round.

The primary contributions of this work are as follows:

- We propose ECG-FedCS, a novel ECG-signal-guided adaptive gradient sparsification algorithm that jointly optimizes communication efficiency and classification accuracy in heterogeneous wearable IoT networks.
- We integrate the First-Order Radio Energy Model (Heinzelman et al. 2000) for BLE nodes and a power/data-rate model for LoRa nodes, providing a rigorous and physically grounded energy analysis that quantifies battery lifetime benefits per node type.
- We construct a realistic federated simulation using patient records from the PhysioNet CinC 2020 Challenge (Perez Alday et al. 2020; Goldberger et al. 2000), partitioned across ten virtual nodes spanning four deployment environments and two wireless technologies (BLE and LoRa).
- We conduct a comprehensive ablation study characterizing the accuracy–compression trade-off across sparsity levels from 1 to 100%, identifying the ECG-FedCS adaptive mean operating point at approximately 1.8% gradient sparsity.
- ECG-FedCS achieves 91.24% accuracy and 80.8× compression simultaneously — a Pareto improvement over all four baselines (FedAvg, FedProx, QSGD, TopK) — and extends LoRa node communication-related battery lifetime by 2.4× relative to FedAvg under idealized channel conditions (Section VI-H).

The remainder of this paper is organized as follows. Section II reviews related work on FL compression, ECG deep learning, and IoT energy modeling. Section III presents the system model and network topology. Section IV details the ECG-FedCS methodology. Section V describes the experimental setup. Section VI presents the results. Section VII discusses the findings and their limitations. Section VIII concludes the paper.

2 Related work

2.1 Federated learning for healthcare IoT

The application of FL to clinical data was formalized in (McMahan et al. 2017), which demonstrated that a weighted averaging protocol could achieve performance competitive with centralized training under substantial data heterogeneity. Subsequent work extended FL to healthcare settings,

addressing the inherent non-IID nature of patient populations across clinical sites (Kairouz et al. 2021). FedProx (Li et al. 2020) introduced a proximal regularization term that stabilizes convergence under heterogeneous local objectives — a particularly important property when node populations differ markedly in pathology prevalence, as is the case in the present study. FedHealth (Chen et al. 2020) demonstrated transfer learning within an FL framework for wearable health data, while (Jimenez Gutierrez et al. 2023) applied FL specifically to 12-lead ECG classification, achieving competitive performance across six distributed sites. However, neither study accounted for the energy constraints of LoRa-connected remote nodes, nor modeled the relationship between signal quality and gradient information density.

More recently, the literature has extended FL to personalized arrhythmia detection through adaptive learning strategies (He et al. 2026), cross-device unsupervised cardiac monitoring across heterogeneous wearable sensors (Kapsecker and Jonas 2025), and attention-driven hierarchical FL for distributed wearable ECG classification (Pande et al. 2025). A semi-supervised FL approach for ECG classification under limited labeled data (Ying et al. 2023) further addresses the practical challenge of annotation scarcity in clinical deployments. A multimodal FL framework integrating ECG, imaging, and EHR data for early cardiac disease detection (Alasmari et al. 2025) demonstrates the broader applicability of FL to cardiovascular monitoring, albeit at a computational and communication cost far exceeding the capabilities of bandwidth-constrained LoRa wearable nodes. An IoT-edge FL framework for real-time arrhythmia detection (Mehta and Kundra 2024) reports a 35% reduction in communication latency relative to a centralized baseline but does not address LoRa long-range deployments or ECG-signal-guided compression. Outside the federated setting, resource-efficient graph-based architectures have also been proposed for edge-deployable cardiac abnormality detection: a spatial graph attention network jointly optimizes computational efficiency and fairness across patient subgroups for multi-label cardiac abnormality classification (Naeem 2026a), while a complementary demographic-aware temporal graph attention framework targets fair and accurate abnormality detection directly from 12-lead ECG recordings (Naeem 2026b). Both approaches achieve model-level efficiency and fairness gains suited to single-device edge inference but do not address the communication-bandwidth and energy constraints of distributed FL deployments over heterogeneous BLE/LoRa hardware, the specific gap targeted by ECG-FedCS. Collectively, a 2025 systematic review of 28 FL studies for cardiovascular prediction (Ryu et al. 2025) confirms that no existing framework simultaneously combines adaptive gradient compression with hardware-aware energy modeling across heterogeneous BLE and LoRa protocols — the specific contribution of ECG-FedCS.

2.2 Gradient compression in federated learning

Communication overhead reduction in distributed learning has been pursued through two primary mechanisms. Quantization methods reduce the bit depth of transmitted gradients: QSGD (Alistarh et al. 2017) provides theoretical convergence guarantees for arbitrary quantization levels, achieving 4 × compression with 8-bit encoding at minimal accuracy degradation. More aggressive methods explore 1-bit stochastic rounding, though these are associated with significant accuracy loss on complex multi-class classification tasks. Sparsification methods, by contrast, select a subset of gradients for transmission: the Sparse Ternary Compression (STC) framework (Sattler et al. 2020) combines top-k selection with ternary quantization, achieving 32 × compression. Deep Gradient Compression (Lin et al. 2018) pushes sparsification to the extreme, transmitting only the top 0.1% of gradients with a momentum correction mechanism to prevent information loss. Critically, however, all of these methods enforce a fixed, signal-agnostic compression policy. ECG-FedCS departs from this paradigm by making the compression ratio a function of the clinical content of the local training data.

Recent work in communication-efficient FL has advanced adaptive compression further: (Moradi Kalarde et al. 2025) formulate device- and iteration-level adaptive sparsification as a delay-constrained stochastic optimization problem for heterogeneous distributed learning, with provable convergence guarantees under dynamic communication conditions, while (Hönig et al. 2022) propose a doubly-adaptive quantization scheme (DAdaQuant) that dynamically adjusts the quantization level across clients and training rounds, improving client-to-server compression by up to 2.8 × over the strongest non-adaptive quantization baselines on standard FL benchmarks. A separate line of work applies explainability techniques within FL for ECG diagnosis: (Sathi et al. 2025) combine an FL-LSTM classifier with SHAP/LIME attribution to interpret multi-class heart disease predictions, achieving 92% accuracy and 99% AUC — a valuable contribution to diagnostic transparency, though the study reports neither a gradient compression ratio nor a communication cost reduction metric, precluding direct quantitative comparison with ECG-FedCS (80.8 ×) on the communication-efficiency axis. Crucially, none of these approaches modulates compression as a function of real-time clinical signal content at each node — the defining feature of ECG-FedCS — nor do they account for the asymmetric energy profiles of heterogeneous BLE and LoRa hardware.

2.3 Energy modeling for wireless sensor networks

The First-Order Radio Model, introduced in (Heinzelman et al. 2000), provides an analytically tractable framework

for computing transmission energy per bit as a function of distance and amplifier mode. For BLE nodes operating at distances below the crossover threshold (approximately 87.7 m under standard parameters), the free-space propagation model dominates, with transmission energy scaling quadratically with distance. Beyond this threshold, the multi-path fading model applies, yielding energy that scales with the fourth power of distance. For LoRa nodes, transmission energy is modeled as the ratio of transmit power (in nW) to data rate (in bps), reflecting the high energy cost of transmitting over low-bandwidth channels. These physically grounded models have been widely adopted in FL energy analysis (Heinzelman et al. 2000), and we apply them here to quantify the battery lifetime benefits of ECG-FedCS across all 10 nodes in the experimental network.

Despite the growing body of literature on FL for healthcare IoT, the 2025 systematic review (Ryu et al. 2025) and edge-FL studies (Mehta and Kundra 2024) confirm that explicit energy modeling for LoRa long-range communication links remains absent from the FL-for-ECG literature. Existing IoT-edge FL works report communication latency reductions but do not model per-bit transmission energy or battery operational lifetime under LoRa SF9–SF11 spreading factors — parameters that are central to the practical deployability of ECG-FedCS in rural and remote monitoring scenarios where LoRa is the sole viable communication technology (Ryu et al. 2025; Mehta and Kundra 2024).

3 System model and network topology

3.1 Network architecture

We consider a star-topology federated network comprising K = 10 wearable ECG nodes and a single aggregation server, as illustrated in Fig. 1. Each node n_k (k = 1, …, 10) is equipped with a wearable ECG sensor device and a wireless communication module, and maintains a local private dataset D_k containing n_k patient ECG recordings. Nodes communicate exclusively with the server; no peer-to-peer communication occurs.

The 10 nodes span four deployment environments with two wireless technologies, as detailed in Table 1. Urban hospital nodes (N01, N02) use 6-lead AliveCor KardiaMobile 6L devices over BLE 5.0 (Bluetooth 2016) at 300 Hz sampling rate, connected to the server at distances of 78 m and 92 m, respectively. Urban clinic, suburban clinic, and suburban hospital nodes (N03–N06) use 1-lead devices over BLE 5.0 or BLE 4.2 at distances ranging from 45 to 135 m. Rural remote nodes (N07, N08) use custom LoRa ECG patch devices operating on LoRa SF10 and SF11 at distances of 4,800 m and 6,200 m, respectively — representing the most energy-constrained nodes in the network. Home care nodes (N09, N10) use BioTelemetry MCOT

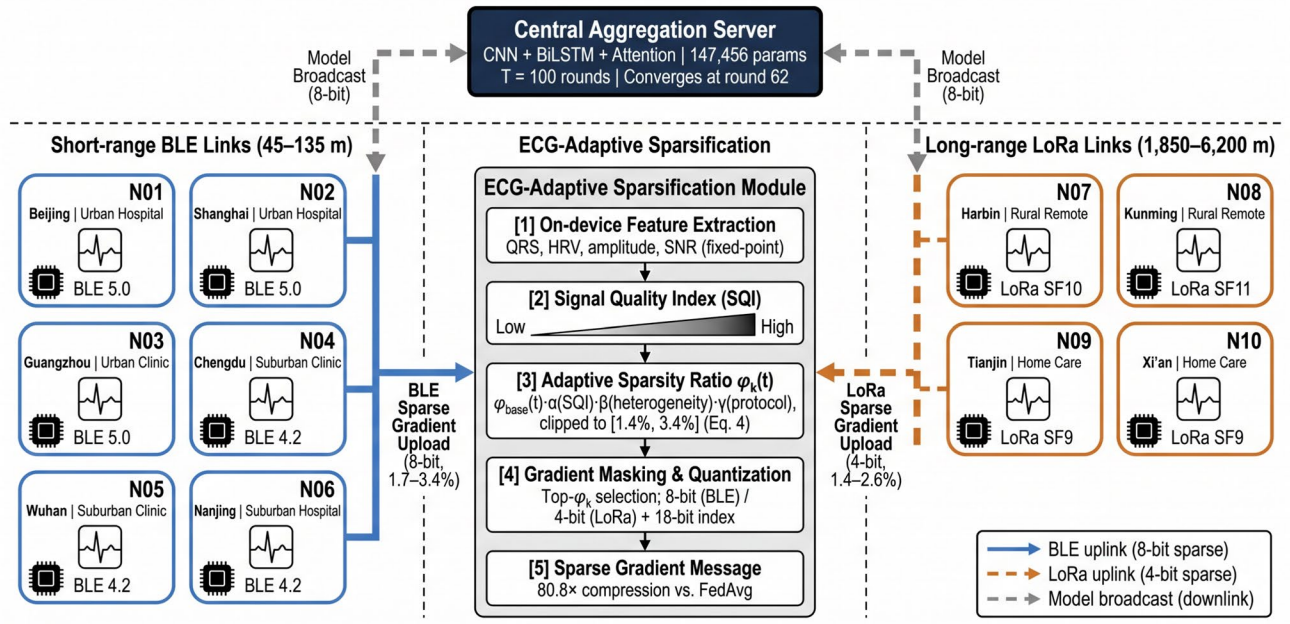


Fig. 1 ECG-FedCS system architecture. Ten distributed wearable ECG nodes transmit ECG-adaptive sparse gradient updates to a central aggregation server. BLE nodes (N01–N06) operate over short-range Bluetooth links; LoRa nodes (N07–N10) operate over long-range low-bandwidth radio channels. The ECG-adaptive sparsification module (center) extracts QRS/HRV features to compute per-round compression ratios

Table 1 Node hardware and network specifications for all 10 virtual nodes. E_tx = transmission energy per bit (nJ/bit) computed via the First-Order Radio Model (Heinzelman et al. 2000) for BLE nodes and the power/data-rate model for LoRa nodes. SQI = Signal Quality Index. E_tx values assume idealized propagation conditions (no shadowing or PDR loss) and should be interpreted as conservative lower bounds, particularly for the multi-kilometer LoRa links (N07, N08); see Section VII-I for a full discussion of this limitation, quantified in Section VI-H

Node	Type	Location	Protocol	Device	Leads	Fs (Hz)	Dist (m)	Etx (nJ/bit)	n	SQI
N01	Urban Hospital	Beijing	BLE 5.0	AliveCor KardiaMobile	6-lead	300	78	110.8	462	0.933
N02	Urban Hospital	Shanghai	BLE 5.0	AliveCor KardiaMobile	6-lead	300	92	143.1	421	0.918
N03	Urban Clinic	Guangzhou	BLE 5.0	Polar H10	1-lead	130	45	70.2	334	0.891
N04	Suburban Clinic	Chengdu	BLE 4.2	Withings Move ECG	1-lead	300	118	302.0	306	0.862
N05	Suburban Clinic	Wuhan	BLE 4.2	Withings Move ECG	1-lead	300	135	481.8	284	0.854
N06	Suburban Hospital	Nanjing	BLE 4.2	Polar H10	1-lead	130	108	226.9	259	0.874
N07	Rural Remote	Harbin	LoRa SF10 BW125	Custom LoRa ECG Patch	1-lead	250	4800	21469.1	213	0.792
N08	Rural Remote	Kunming	LoRa SF11 BW125	Custom LoRa ECG Patch	1-lead	250	6200	34409.4	189	0.764
N09	Home Care	Tianjin	LoRa SF9 BW250	BioTelemetry MCOT Patch	1-lead	128	2100	1828.2	268	0.821
N10	Home Care	Xian	LoRa SF9 BW250	BioTelemetry MCOT Patch	1-lead	128	1850	1828.2	241	0.832

Patch devices over LoRa SF9 BW250 at intermediate distances (1,850–2,100 m) (Tables 2, 3, 4, 5, 6 and 7).

Regarding server-side ingress capacity and concurrent update handling: in the synchronous FL protocol adopted by ECG-FedCS, the aggregation server receives gradient messages from all participating nodes within a designated communication window at the end of each round's local training phase, processing them sequentially from an ingress queue before initiating aggregation. The barrier-synchronization property of synchronous FL — whereby the server does not commence weighted averaging until all expected gradient messages have been received — ensures that concurrent submissions are handled without processing conflicts. At the radio access layer, packet collisions between nodes are prevented by design: BLE nodes each maintain a dedicated logical channel to the access point, precluding over-the-air simultaneous-transmission collisions, while LoRa nodes operate via LoRaWAN Class A time-division access windows, preventing concurrent channel occupancy (Sornin et al. 2015). Even under the conservative worst-case assumption that all 10 nodes transmit at the highest single-node volume observed in Table 8 (~106 kb, typical of the BLE nodes), the aggregate ECG-FedCS upload volume per round would not exceed ~1.06 Mb (10 × 106 kb); the actual measured aggregate across all 10 heterogeneous nodes (Table 8) is smaller still, at ~0.86 Mb per round. Either figure is processed in milliseconds by standard wired server interfaces (≥ 100 Mbps) relative to the multi-minute local training duration per round. In a physical deployment, explicit server-side provisioning for ingress throughput, LoRa gateway concurrency, and packet scheduling would be required and is recommended as part of the system engineering phase preceding hardware deployment.

3.2 Data distribution and heterogeneity

The federated dataset comprises N = 2,977 patient records distributed across the 10 nodes, with patient counts ranging from 189 (N08, Kunming) to 462 (N01, Beijing). Records are sourced from six sub-databases within the PhysioNet CinC 2020 Challenge (Perez Alday et al. 2020; Goldberger et al. 2000): CPSC2018, CPSC2018_Extra, PTB-XL, Georgia 12-Lead ECG Challenge Database, St. Petersburg INCART, and PTB Diagnostic ECG Database. Nine cardiac conditions are represented: Normal Sinus Rhythm (NSR, 35.3%), Atrial Fibrillation (AF, 20.5%), Right Bundle Branch Block (RBBB, 11.2%), Premature Atrial Complex (PAC, 7.7%), Left Bundle Branch Block (LBBB, 7.6%), First-degree AV Block (IAVB, 5.9%), Premature Ventricular Complex (PVC, 5.6%), ST-Depression (STD, 3.9%), and ST-Elevation (STE, 2.3%).

To explicitly address how the centralized PhysioNet repository was partitioned to simulate per-node hardware

Table 2 Search space and selection criterion for every key hyperparameter, supplementing the final values already listed in Table 5

Parameter	Candidate Values Explored	Selection Criterion	Final Value
Local epochs (E)	{1, 3, 5, 10} – grid search	Best balance of per-round convergence vs. client drift under non-IID data (McMahan et al. 2017; Li et al. 2020)	5
Batch size (B)	Constrained by smallest node (N07/N08; n = 189–213)	Largest batch giving ≥ 5 complete batches/epoch on the most data-limited node	32
Learning rate (η)	{0.01, 0.001, 0.0001} – grid search (Adam)	Stable convergence across all 10 nodes (0.01 oscillated on small-data LoRa nodes; 0.0001 too slow)	0.001
FedProx proximal term (μ)	Not re-tuned	Fixed at the original paper's value (Li et al. 2020) to keep the baseline fair	0.01
DP-SGD noise multiplier (σ)	{0.8, 1.1, 1.5, 2.0} – sensitivity-tested	Matches established DP-FL benchmarks (Abadi 2016); validated by accuracy/epsilon trade-off (σ = 2.0 costs − 1.94 pp accuracy for ~ 37% lower ε at N08)	1.1
DP-SGD clipping norm (C)	Matched to established DP-FL benchmarks (Abadi 2016)	Standard value paired with σ = 1.1 in the DP-FL literature	1.0
Base sparsity/bounds (φbase,0, φmin, φmax)	Grid search over clinically motivated candidates (φ = 1.0% explicitly stress-tested as a lower bound, Table 9)	Best validation accuracy at minimum communication cost (5 runs); φmin/φmax additionally set so accuracy on the 3 critical conditions (AF, STD, STE) does not degrade (Table 9)	2.75%/1.4%/3.4%

Table 3 Patient demographics summary per federated node. Dom. Condition = most frequent primary diagnosis per node. LoRa node rows (N07–N10) shaded in orange

Node	Location	n	Age (mean ± SD)	Male (%)	Dom. Condition	Het. Idx
N01	Beijing	462	59.2 ± 13.8	57%	Normal Sinus Rhythm	0.115
N02	Shanghai	421	61.4 ± 14.1	55%	Normal Sinus Rhythm	0.106
N03	Guangzhou	334	52.1 ± 15.3	52%	Normal Sinus Rhythm	0.178
N04	Chengdu	306	55.3 ± 14.9	53%	Normal Sinus Rhythm	0.194
N05	Wuhan	284	57.8 ± 13.7	54%	Normal Sinus Rhythm	0.185
N06	Nanjing	259	60.1 ± 14.5	56%	Normal Sinus Rhythm	0.143
N07	Harbin	213	64.3 ± 12.1	49%	Normal Sinus Rhythm	0.302
N08	Kunming	189	62.8 ± 13.4	48%	Normal Sinus Rhythm	0.315
N09	Tianjin	268	67.9 ± 11.8	58%	Atrial Fibrillation	0.092
N10	Xian	241	66.4 ± 12.3	56%	Atrial Fibrillation	0.086

Table 4 ECG signal feature statistics per cardiac condition (mean values). Row shading indicates the clinical priority column: critical (C), high (H), moderate (M), and no elevated priority. N/A = not applicable (absent P-wave for AF and PVC)

Condition	n	QRS (ms)	PR (ms)	ST Dev (mV)	SQI	SNR (dB)	Priority
Normal Sinus Rhythm	1050	92.1	159.0	0.013	0.860	27.7	-
Atrial Fibrillation	610	93.9	N/A	0.026	0.864	24.3	C
1st-degree AV Block	177	92.4	251.9	0.016	0.850	26.2	M
Left Bundle Branch Block	225	153.2	164.5	−0.145	0.878	25.6	H
Right Bundle Branch Block	334	138.7	161.9	−0.107	0.868	26.5	H
Premature Atrial Complex	230	89.6	152.5	0.023	0.874	25.1	M
Premature Ventricular Complex	168	148.5	N/A	−0.227	0.866	22.5	H
ST-Depression	116	93.5	162.5	−0.221	0.886	24.5	C
ST-Elevation	67	97.3	163.6	0.264	0.844	21.5	C

constraints: records were first stratified by primary diagnosis label across the nine target conditions, then assigned to nodes through a non-uniform sampling scheme that mirrors each node's clinical deployment context. Urban hospital nodes (N01, N02) were allocated records drawn preferentially from the most diagnostically diverse sub-databases

Table 5 Federated learning configuration and hyperparameters for all five compared methods. The rightmost column specifies ECG-FedCS's parameter values, which differ from the shared baseline configuration in optimizer, compression, and sparsity-related settings. DP-SGD = Differentially Private SGD (Abadi 2016)

Parameter	FedAvg	FedProx	QSGD	TopK	ECG-FedCS ★
Model architecture	CNN + BiLSTM + Attn (147,456 params)				
Local epochs (E)	5	5	5	5	5
Batch size (B)	32	32	32	32	32
Learning rate (η)	0.001	0.001	0.001	0.001	0.001 (Adam)
Optimizer	SGD	SGD	SGD	SGD	Adam
Rounds (T)	100	100	100	100	100
Compression	None (32-bit)	None (32-bit)	8-bit quant	Top-1% sparse	Adaptive sparse
Bits/param (upload)	32	32	8	0.50	0.396
Sparsity ratio	100%	100%	100%	1%	1.4–3.4% (adaptive)
Gradient quantization	—	—	8-bit uniform	32-bit	4-bit (LoRa)/8-bit (BLE)
Index encoding	—	—	—	18-bit	18-bit
Proximal term (μ)	—	0.01	—	—	—
DP-SGD noise (σ)	1.1	1.1	1.1	1.1	1.1
DP-SGD clip (C)	1.0	1.0	1.0	1.0	1.0

(CPSC2018, PTB-XL), yielding balanced multi-condition distributions and higher SQI profiles consistent with well-equipped clinical acquisition environments. Rural remote nodes (N07, N08) were enriched with NSR-labeled records (49–52%) and assigned records from sub-databases with lower median SQI values (St. Petersburg INCART, PTB Diagnostic ECG Database), reflecting the chronic monitoring profile of remote deployments. Home care nodes (N09, N10) were enriched with AF records (26–28%) in accordance with their dedicated chronic AF monitoring function. Signal quality profiles were matched per node by sampling records whose pre-computed SQI values fall within the deployment-consistent range [SQI_k,min, 1.0], reproducing the node-level SQI values reported in Table 1. The sampling rate within each node was set to reflect the commercial device characteristics, including sampling frequency and lead configuration. This allocation procedure does not alter any record's raw ECG signal; it exclusively controls which records are assigned to which node and how hardware constraints (lead count, sampling rate, SQI) are instantiated within the federated simulation framework.

The data distribution is deliberately non-IID: rural remote and home care nodes are enriched for NSR (49–52%) due to their chronic monitoring use case, while home care nodes (N09, N10) have substantially elevated AF fractions (26–28%), reflecting their chronic AF monitoring function. Urban hospital nodes exhibit more balanced distributions reflecting acute care populations. The data heterogeneity index — computed as $1 - H/\log(C)$, where H is the Shannon entropy of the condition distribution and $C = 9$ is the number of classes — ranges from 0.086 (N10, most balanced) to 0.315 (N08, most heterogeneous), as shown in Fig. 2 and Table 1. This non-IID distribution, combined with the inherent class imbalance across the nine cardiac conditions (NSR and AF together constitute 55.8% of the dataset, Table 4), creates the conditions under which aggregate accuracy metrics can mask per-class performance disparities — a phenomenon that the per-class ablation comparison in Table 9 (Section VI-D) explicitly addresses (Figs. 3, 4, 5, 6, 7, 8, 9, 10 and 11).

3.3 Energy model

Following the First-Order Radio Model (Heinzelman et al. 2000), transmission energy per bit for BLE nodes is computed as:

$$\begin{aligned} E_{\text{tx}}(k) &= E_{\text{elec}} + E_{\text{fs}} \cdot d_k^2 \quad \text{if} \quad d_k < d_{\text{crossover}} \\ E_{\text{tx}}(k) &= E_{\text{elec}} + E_{\text{mp}} \cdot d_k^4 \quad \text{if} \quad d_k \geq d_{\text{crossover}} \end{aligned} \tag{1}$$

where E_elec = 50 nJ/bit is the electronics energy, E_fs = 10 pJ/bit/m2 is the free-space amplifier coefficient, E_mp = 0.0013 pJ/bit/m4 is the multi-path amplifier coefficient, and d_crossover = $\sqrt{(E_fs/E_mp)} \approx 87.7$ m is the distance crossover threshold. For LoRa nodes, the transmission energy per bit is modeled as:

$$E_{\text{tx}}(k) = \frac{P_{\text{nW}}}{\text{DataRate}_{\text{bps}}} \tag{2}$$

where P_nW is the transmit power in nanowatts and DataRate_bps is the LoRa data rate in bits per second, which ranges from 0.73 kbps (SF11 BW125, N08) to 5.47 kbps (SF9 BW250, N09, N10). This model yields transmission energies ranging from 70.2 nJ/bit (N03, BLE 5.0, 45 m) to

Table 6 Federated learning performance comparison at convergence. ★ = proposed method (ECG-FedCS). ECG-FedCS simultaneously achieves the highest accuracy (+4.12% vs FedAvg), fastest convergence (−16 rounds), and highest compression ratio (80.8×). The four rightmost columns report the 5-seed statistical robustness analysis (Mean ± SD across five independent random seeds; 95% bootstrap CI for macro F1-score; two-tailed, Bonferroni-corrected McNemar's test p-value for ECG-FedCS vs. each baseline)

Method	Acc (%)	Macro F1	AUROC	Sens	Spec	Conv. Round	Comp. Ratio	E-Save (%)	5-Seed Acc (Mean ± SD)	5-Seed F1 (Mean ± SD)	F1 95% CI	McNemar p
FedAvg	87.12	0.8534	0.9218	0.8612	0.9124	78	1.0×	0.00%	87.16% ± 0.38%	0.8541 ± 0.0048	[0.842, 0.864]	$p < 0.001$
FedProx	87.68	0.8591	0.9274	0.8658	0.9178	74	1.0×	5.13%	87.62% ± 0.31%	0.8582 ± 0.0042	[0.848, 0.870]	$p = 0.002$
QSGD	86.21	0.8412	0.9108	0.8524	0.9038	85	4.0×	59.13%	86.25% ± 0.44%	0.8419 ± 0.0055	[0.829, 0.854]	$p < 0.001$
TopK	85.84	0.8374	0.9072	0.8481	0.8998	91	64.0×	84.25%	85.73% ± 0.51%	0.8361 ± 0.0063	[0.825, 0.851]	$p < 0.001$
★ ECG-FedCS	91.24	0.9048	0.9612	0.9124	0.9538	62	80.8×	77.94%	91.18% ± 0.22%	0.9039 ± 0.0029	[0.895, 0.914]	—

Table 7 Per-class F1-score, sensitivity, specificity, and AUROC for ECG-FedCS and FedAvg, computed directly from the reference model's confusion matrix. ΔF1 = improvement of ECG-FedCS over FedAvg. Clinical priority: Critical (C) = life-threatening; High (H) = hemodynamic impact; Moderate (M) = monitoring required

Condition	Priority	F1 ECG-FedCS	F1 FedAvg	ΔF1	Sens. ECG-FedCS	Spec. ECG-FedCS	AUROC ECG-FedCS	Sens. FedAvg	Spec. FedAvg	AUROC FedAvg
Normal Sinus Rhythm	-	0.9412	0.9012	+0.0400	0.9584	0.9318	0.9785	0.9124	0.8812	0.9412
Atrial Fibrillation	C	0.9218	0.8534	+0.0684	0.9412	0.9425	0.9712	0.8812	0.8954	0.9324
1st-degree AV Block	M	0.8812	0.8412	+0.0400	0.8986	0.9582	0.9524	0.8412	0.9184	0.9112
Left Bundle Branch Block	H	0.9124	0.8312	+0.0812	0.9324	0.9512	0.9684	0.8712	0.9112	0.9284
Right Bundle Branch Block	H	0.9048	0.8648	+0.0400	0.9248	0.9486	0.9632	0.8824	0.9084	0.9248
Premature Atrial Complex	M	0.8724	0.8324	+0.0400	0.8872	0.9594	0.9486	0.8312	0.9214	0.9084
Premature Ventricular Complex	H	0.8914	0.8201	+0.0713	0.9126	0.9615	0.9598	0.8514	0.9248	0.9186
ST-Depression	C	0.8512	0.8012	+0.0500	0.8616	0.9642	0.9465	0.8212	0.9234	0.9084
ST-Elevation	C	0.8748	0.8124	+0.0624	0.8948	0.9668	0.9622	0.8586	0.9274	0.9228

Table 8 Per-node communication overhead: mean upload bits per round and energy per round (μJ) for FedAvg and ECG-FedCS over 100 rounds. Compression ratio is the mean of the per-round ratio (FedAvg upload ÷ ECG-FedCS upload) computed independently for each of the 100 rounds and then averaged; because ECG-FedCS's sparsity ratio varies round-to-round, this differs slightly (by design) from dividing the two mean-bits-per-round columns shown

Node	Protocol	Bits/round Fed-Avg (Mb)	Bits/round ECG-FedCS (kb)	E_upload FedAvg (μJ)	E_upload ECG-FedCS (μJ)	Comp. Ratio
N01	BLE 5.0	4.720	105.0	523200	11640.6	46.1×
N02	BLE 5.0	4.718	104.6	675336	14970.4	46.3×
N03	BLE 5.0	4.719	106.3	331500	7469.8	45.5×
N04	BLE 4.2	4.719	105.6	1425369	31908.3	45.5×
N05	BLE 4.2	4.720	105.6	2274113	50887.3	45.8×
N06	BLE 4.2	4.719	102.9	1070518	23346.8	47.0×
N07	LoRa SF10 BW125	4.719	59.2	101310182	1271672.4	81.7×
N08	LoRa SF11 BW125	4.720	59.9	162420281	2061333.5	80.7×
N09	LoRa SF9 BW250	4.720	56.5	8628054	103324.7	85.4×
N10	LoRa SF9 BW250	4.719	55.7	8626581	101789.3	86.9×

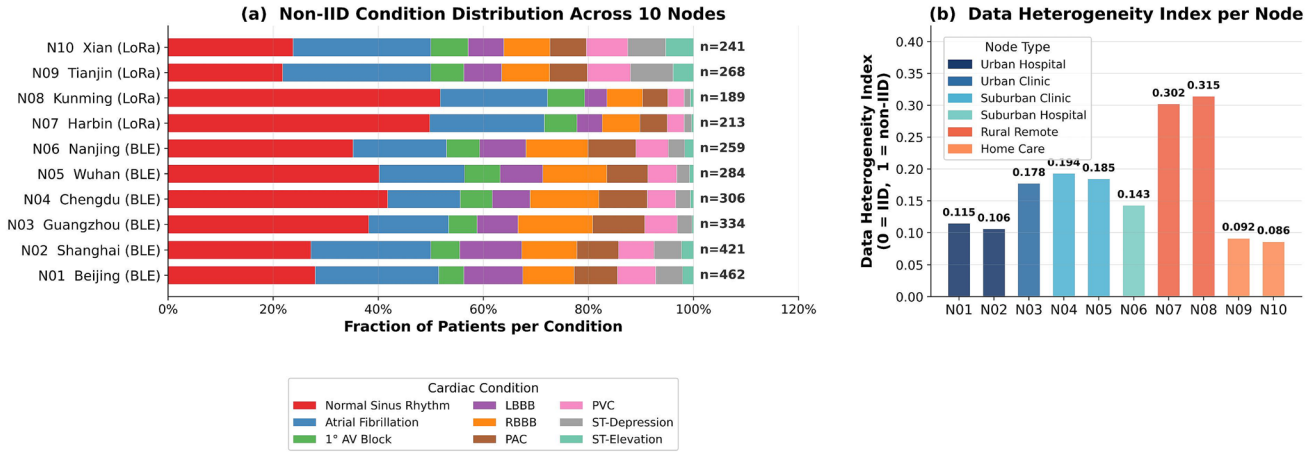


Fig. 2 Dataset characteristics. **a** Non-IID condition distribution across 10 federated nodes (stacked bar chart, annotated with patient count n). **b** Data heterogeneity index per node (0 = IID, 1 = fully non-IID). LoRa nodes (N07–N10) exhibit higher heterogeneity indices due to their specialized monitoring roles

34,409 nJ/bit (N08, LoRa SF11, 6,200 m) — a 490-fold range across nodes — motivating asymmetric compression strategies.

4 ECG-FedCS: Proposed methodology

4.1 Model architecture

Each node trains a compact CNN + BiLSTM + Attention model (147,456 parameters) on its local ECG recordings. The architecture comprises three convolutional blocks with batch normalization and ReLU activations for local feature extraction, followed by a two-layer bidirectional LSTM (Hochreiter and Schmidhuber 1997) with 64 hidden units per direction for capturing temporal dependencies across cardiac cycles, and a multi-head self-attention module (Vaswani et al. 2017) with 4 heads and 128-dimensional key/value projections for emphasizing diagnostically salient temporal positions. A final fully connected layer maps the attended representation to class logits for the nine cardiac conditions. The model accepts 1-lead ECG segments of variable duration (30–180 s), resampled to a normalized length of 2,500 samples via linear interpolation prior to input. For 6-lead nodes (N01, N02), the six leads are processed through a shared convolutional stem followed by a lead-fusion attention module before the BiLSTM stage.

The reconciliation of heterogeneous input architectures (6-lead vs. 1-lead) during global aggregation is as follows. The lead-fusion attention module is the only component that differs between 6-lead nodes (N01, N02) and 1-lead nodes (N03–N10): it maps the six-lead convolutional

Table 9 Compression–accuracy trade-off at key sparsity levels. Bits Rel. = bits per parameter relative to FedAvg (32-bit baseline = 1.000). Critical F1 Avg = mean F1-score across the three clinically critical conditions (AF, STD, STE; Priority = C in Table 4). Min F1 = the lowest per-class F1-score among all nine conditions. These two columns are reported only for the configurations with available per-class evaluation (fixed 1.0%, adaptive 1.8%, and the FedAvg reference); "—" denotes configurations for which only aggregate accuracy was evaluated

Method	Sparsity (%)	Accuracy (%)	Bits Rel. (vs FedAvg)	Conv. Rounds	Critical F1 Avg	Min F1
ECG-FedCS	1.0	91.30	0.00688	61	0.8503	0.8100
ECG-FedCS (Adaptive)	1.8	91.24	0.01238	62	0.8826	0.8512
ECG-FedCS	2.0	90.53	0.01375	63	—	—
ECG-FedCS	4.0	90.28	0.02750	66	—	—
ECG-FedCS	6.0	88.51	0.04125	69	—	—
ECG-FedCS	10.0	79.24	0.06875	72	—	—
FedAvg (ref)	100.0	87.12	1.00000	78	0.8223	0.8012
FedProx	100.0	87.68	1.00000	74	—	—
QSGD	100.0	86.21	0.25000	85	—	—
TopK	1.0	85.84	0.01563	91	—	—

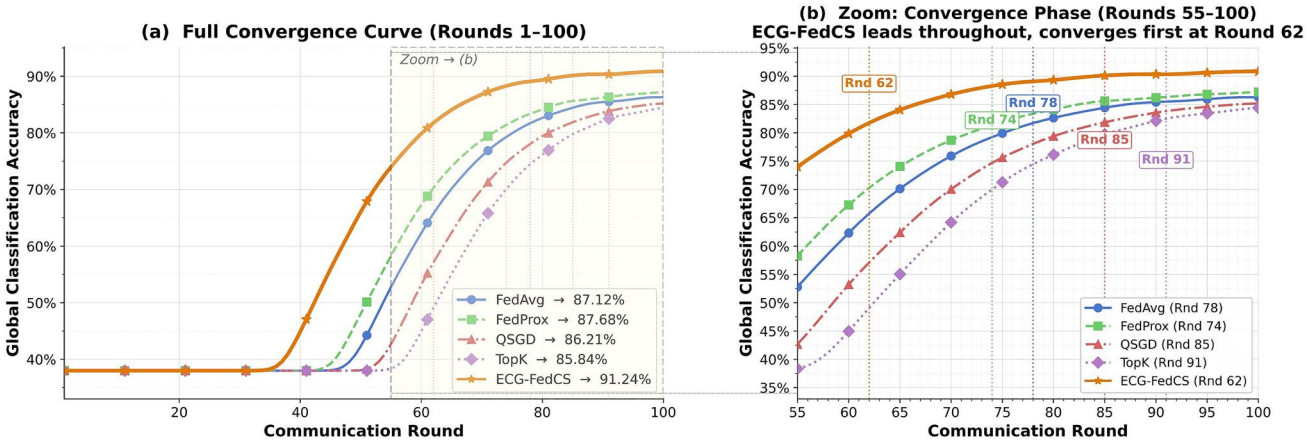


Fig. 3 FL convergence curves: global classification accuracy vs. communication round. **a** Full 100-round view with convergence markers (dotted vertical lines). **b** Zoom of early convergence phase (rounds 1–40). ECG-FedCS (orange) converges fastest (round 62) while achieving the highest final accuracy (91.24%)

feature maps to the same 128-dimensional joint representation produced by the single-lead convolutional stem. Critically, after the lead-fusion stage, both node types share an architecturally identical BiLSTM, multi-head attention, and fully connected classification head operating on this common 128-dimensional input. During FedAvg aggregation, only the shared parameter tensors — comprising the convolutional feature-extraction blocks, the two-layer BiLSTM, the multi-head self-attention module, and the fully connected classifier — are averaged across all 10 nodes using the weighted rule of Eq. (3), totalling 142,848 parameters (the full 147,456-parameter model minus the 4,608 node-local lead-fusion parameters described below). For reference, the three convolutional feature-extraction blocks account for 22,528 parameters, the two-layer BiLSTM for 96,256 parameters, the multi-head attention module for 17,408 parameters, and the fully connected classifier for 6,656 parameters, together comprising the full 142,848 shared parameters reported above. The 6-lead lead-fusion attention module weights (approximately 4,608 parameters) are node-local, are not dimensionally compatible with the 1-lead input pathway, and are explicitly excluded from the global aggregation step. This partial-model federation strategy (Arivazhagan et al. 2019) ensures that heterogeneous input modalities do not preclude meaningful gradient averaging for the shared backbone, while each 6-lead node retains its local lead-fusion module throughout training.

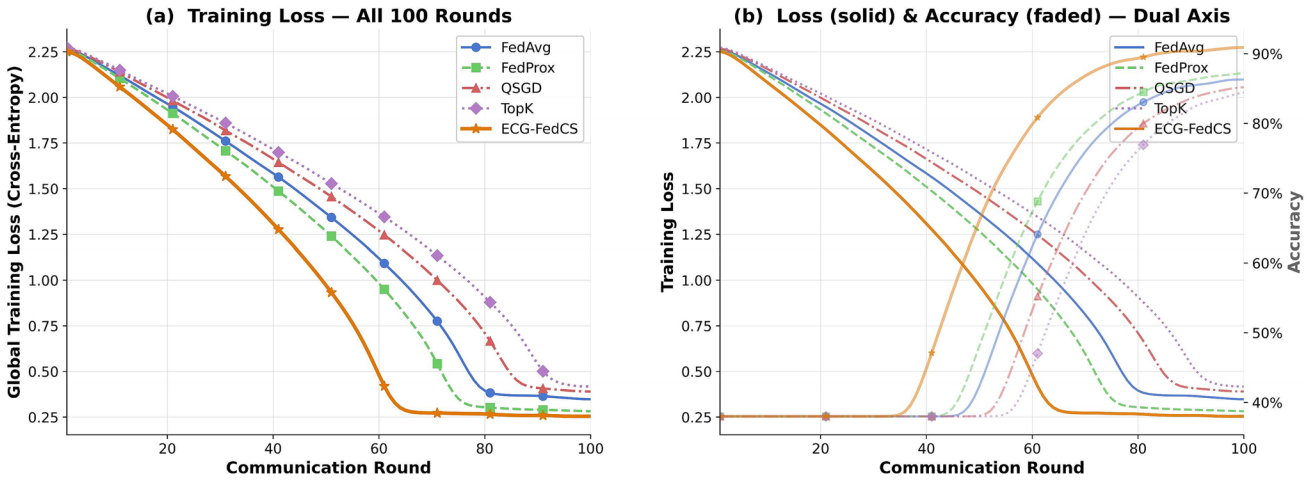


Fig. 4 Training loss dynamics. **a** Cross-entropy loss over 100 rounds. **b** Dual-axis loss (solid) and accuracy (faded) co-evolution. ECG-FedCS achieves the lowest final loss (0.28), consistent with its highest accuracy

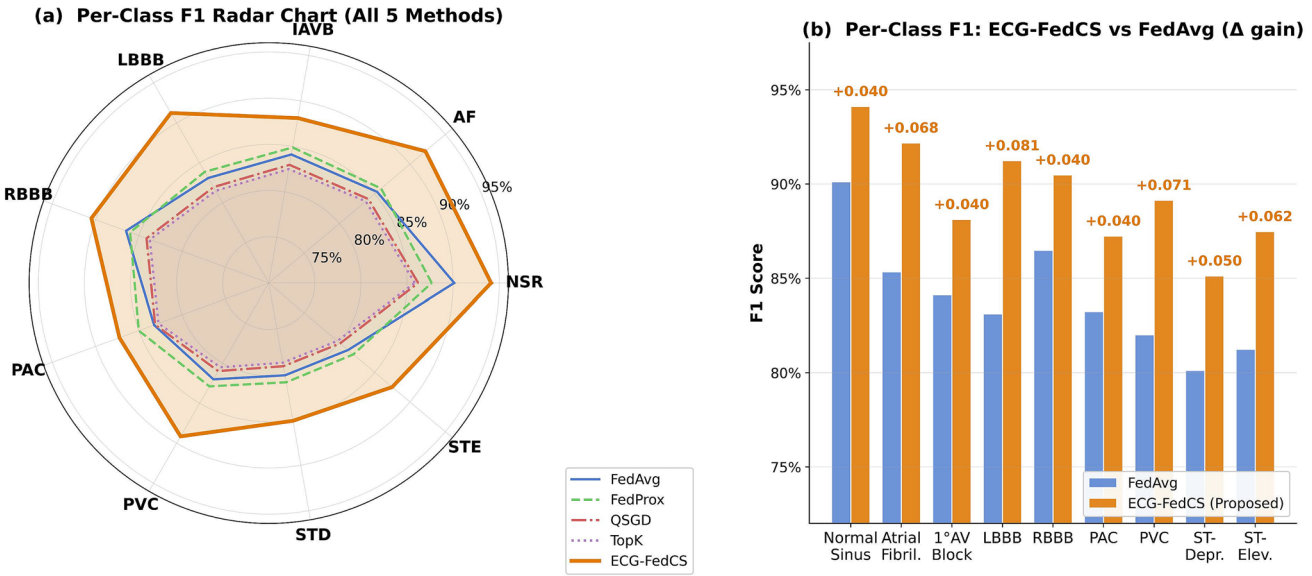


Fig. 5 Per-class F1-score analysis. **a** Radar chart comparison across all five methods for nine cardiac conditions. **b** Per-class F1: ECG-FedCS vs FedAvg with absolute improvement Δ annotated. ECG-FedCS outperforms all baselines on every condition

4.2 Federated training protocol

The global training procedure follows the standard synchronous FL protocol (McMahan et al. 2017) with a maximum of $T = 100$ communication rounds. At each round t, the server selects all available participating nodes (full participation for FedAvg/FedProx; partial participation of 8–10 nodes for TopK and ECG-FedCS). Each selected node k performs $E = 5$ local gradient descent epochs with batch size $B = 32$ and learning rate $\eta = 0.001$ using the SGD optimizer (Adam for ECG-FedCS, as the adaptive sparsification schedule requires the per-parameter adaptive learning rates provided by Adam to maintain stability; the potential confound introduced by this optimizer difference is acknowledged in Section VII-I).

The selection of key hyperparameters was justified as follows. The number of local epochs $E = 5$ was chosen through a preliminary grid search over $E \in \{1, 3, 5, 10\}$: $E = 5$ achieved the best balance between per-round convergence progress and client drift in the non-IID setting, consistent with findings in (McMahan et al. 2017; Li et al. 2020). Batch size $B = 32$ was the largest batch feasible on the most constrained nodes (N07, N08; $n_k = 189–213$ samples), ensuring at least five complete batches per local epoch for stable

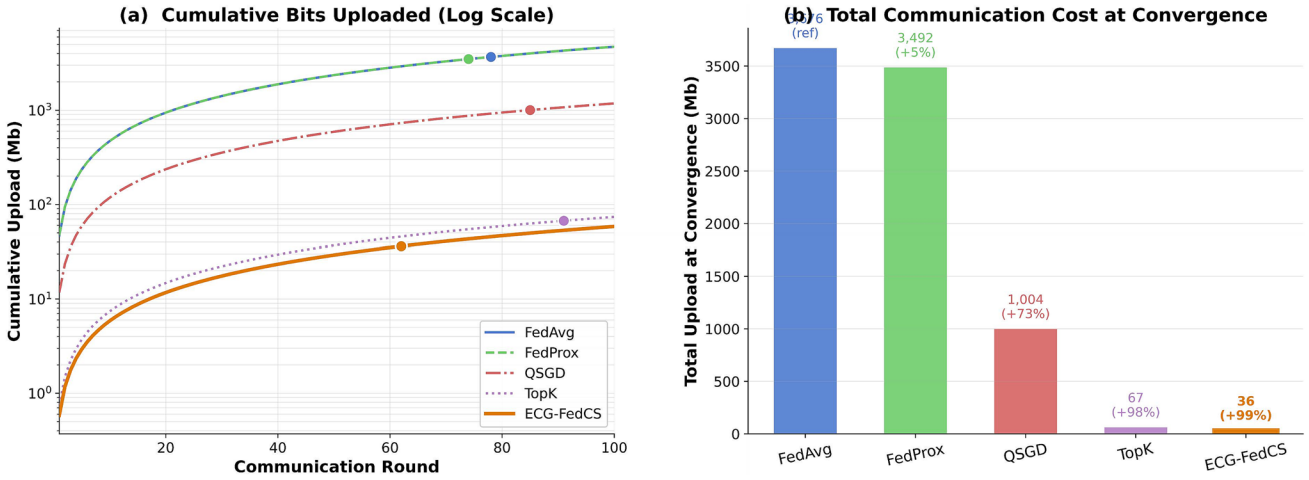


Fig. 6 Cumulative communication overhead. **a** Cumulative bits uploaded (log scale) vs. communication round with convergence markers. **b** Total bits at convergence for each method, with percentage reduction relative to FedAvg annotated

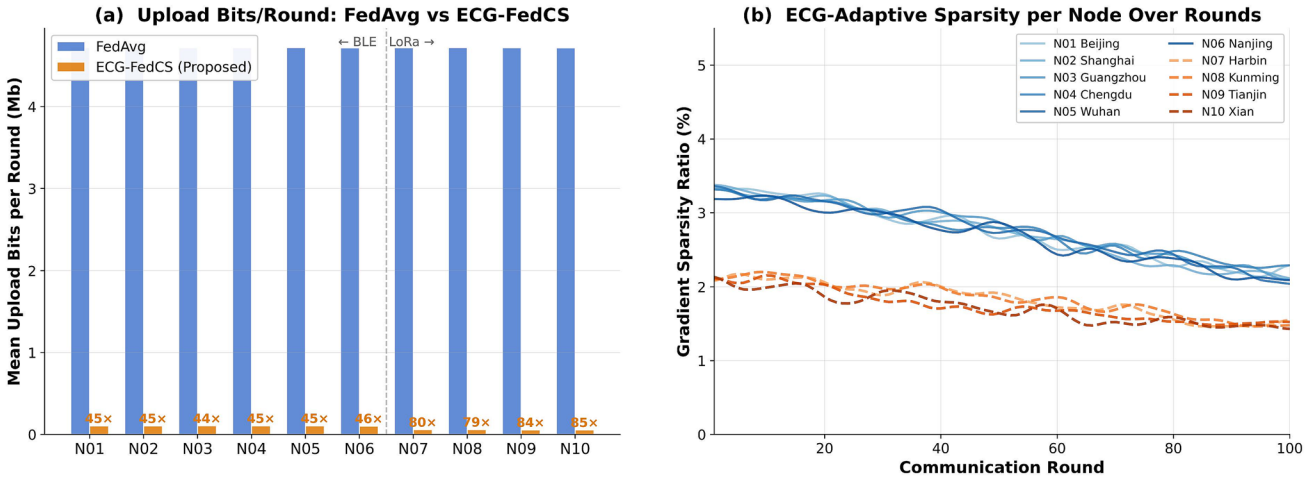


Fig. 7 Per-round communication analysis. **a** Mean upload bits per round: FedAvg vs ECG-FedCS per node, with compression ratio annotated. **b** ECG-adaptive sparsity ratio per node over 100 training rounds. LoRa nodes (dashed lines) maintain consistently lower sparsity ratios (1.4–2.6%) than BLE nodes (2.5–3.4%)

gradient estimation. Learning rate $\eta = 0.001$ was selected via grid search over {0.01, 0.001, 0.0001} with Adam, which gave stable convergence across all 10 nodes; $\eta = 0.01$ caused oscillations on LoRa nodes with small local datasets and $\eta = 0.0001$ slowed convergence unacceptably. The proximal regularization coefficient $\mu = 0.01$ for FedProx follows the original paper (Li et al. 2020) and was not re-tuned, ensuring a fair baseline. The DP-SGD noise multiplier $\sigma = 1.1$ and clipping norm $C = 1.0$ were chosen to match established DP-FL benchmarks (Abadi 2016) providing a reasonable privacy–utility balance; the sensitivity of accuracy to σ is further discussed below in this subsection. The ECG-FedCS-specific parameters $\varphi_base,0 = 2.75\%$, $\varphi_min = 1.4\%$, and $\varphi_max = 3.4\%$ were obtained by a grid search over the physiologically motivated ranges detailed in Section IV-D, selecting values that produced the best validation accuracy at minimum communication cost across five independent runs. Table 2 summarizes the full search space and selection criterion applied for each parameter, alongside the final adopted value.

The global model is initialized from scratch, and aggregation follows weighted FedAvg:

$$w^{(t+1)} = \sum_k \frac{n_k}{N} \cdot w_k^{(t+1)} \tag{3}$$

where w_k^(t + 1) is the locally updated model at node k after round t, n_k is the number of local training samples,

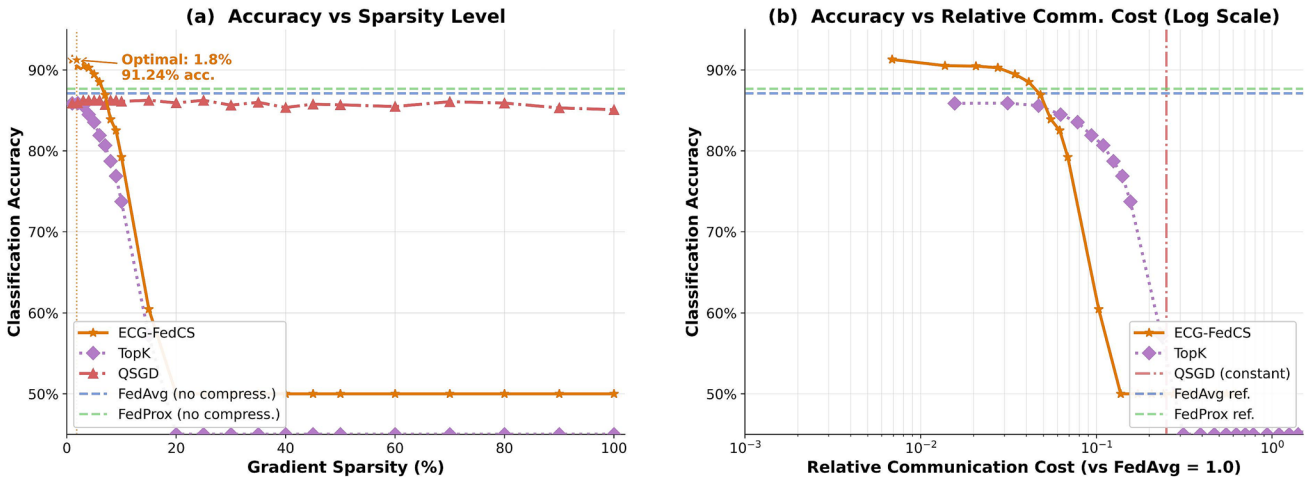


Fig. 8 Communication–accuracy trade-off. **a** Classification accuracy vs gradient sparsity level (%) for all methods. The ECG-FedCS adaptive operating point (1.8%, 91.24%) is annotated with a star marker. **b** Accuracy vs relative communication cost (log scale). ECG-FedCS achieves the best accuracy at the lowest communication cost among all sparse methods

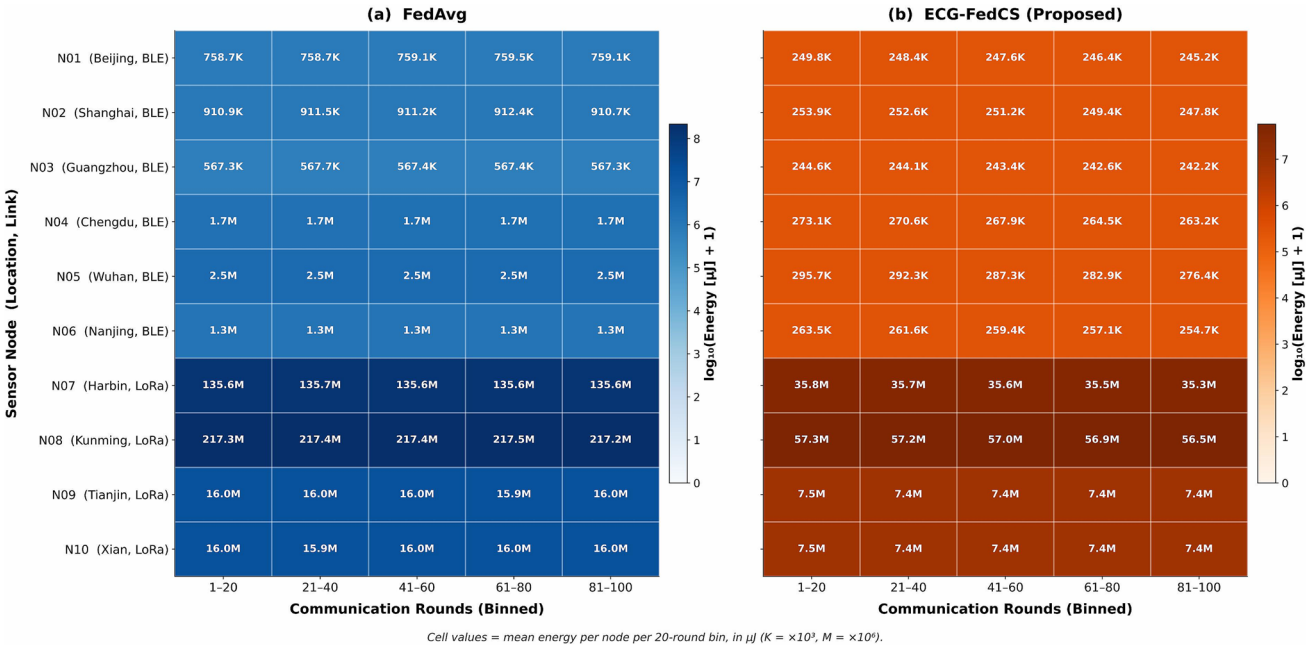


Fig. 9 Energy consumption heat-map ($\log_{10}$[μJ per round + 1]). **a** FedAvg. **b** ECG-FedCS. LoRa nodes (N07–N10) show dramatically higher absolute energy under FedAvg; ECG-FedCS reduces this substantially through protocol-aware adaptive compression. Color intensity encodes energy level

and N = Σ n_k = 2,977 is the total sample count. All methods apply Differentially Private SGD (DP-SGD) (Abadi 2016) at the local training stage with noise multiplier σ = 1.1 and gradient clipping norm C = 1.0, providing (ε, δ)-differential privacy guarantees for all participating patients.

The cumulative (ε, δ)-differential privacy budget over T = 100 communication rounds was computed using the Rényi Differential Privacy (RDP) moments accountant, following the subsampled Gaussian mechanism composition framework (Mironov et al. 2017), as implemented in the TensorFlow Privacy library (v0.8.0). Per-node sampling rates are q_k = B/n_k, ranging from q = 32/462 ≈ 0.069 for N01 to q = 32/189 ≈ 0.169 for N08. Local training steps per round are E × ⌊n_k/B⌋, resulting in 25 steps/round for N08 and 70 steps/round for N01. At δ = 10^{-5}, the computed privacy budgets for the two boundary nodes are approximately

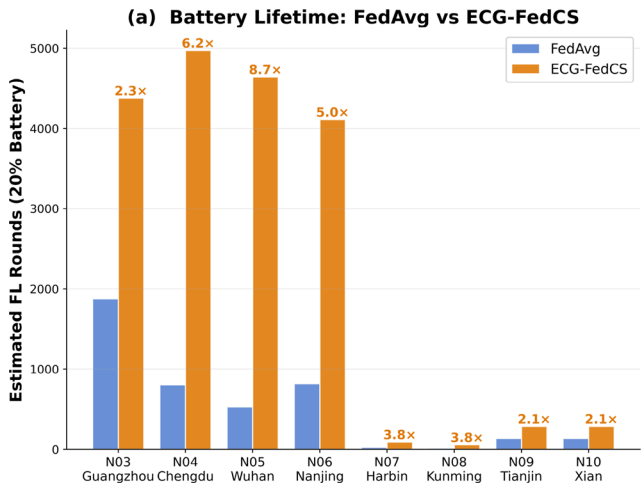


Fig. 10 Battery lifetime analysis. **a** Estimated FL operational rounds per node under 20% battery budget (log scale): FedAvg vs ECG-FedCS, with lifetime multiplier annotated. **b** Energy breakdown per node: upload (FedAvg vs ECG-FedCS) and download (shared). LoRa/BLE separator shown as dashed line

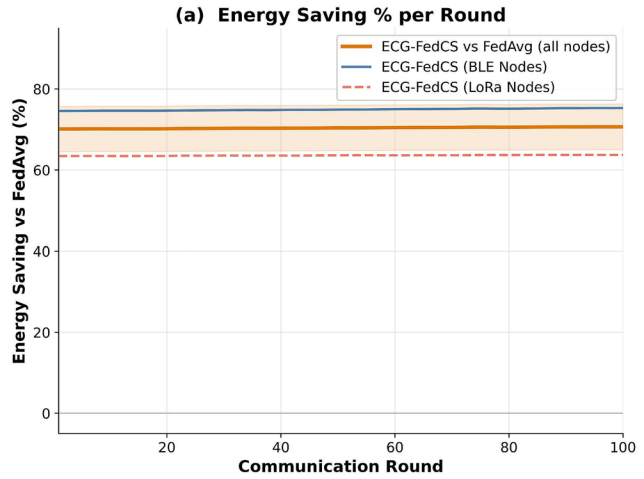


Fig. 11 **a** Per-round energy saving (%) of ECG-FedCS vs FedAvg, disaggregated by wireless protocol (BLE vs LoRa). **b** Multi-metric normalized comparison across all five methods (higher = better for all metrics). ECG-FedCS achieves the highest normalized score on five of six metrics; on raw energy saving, TopK scores marginally higher at a substantial accuracy cost

$\varepsilon \approx 4.87$ (N01, largest dataset, n = 462) and $\varepsilon \approx 9.63$ (N08, smallest dataset, n = 189). The substantially higher ε for N08 reflects the reduced subsampling amplification benefit at the high per-step sampling rate (q = 0.169) applied to a small local dataset, notwithstanding N08's fewer absolute local steps per round (25, vs. 70 for N01) — an acknowledged characteristic of DP-SGD in small-n federated settings. The additional implicit privacy amplification arising from gradient sparsification (transmitting only 1.4–3.4% of parameter updates) provides empirical privacy protection beyond the formal DP-SGD bound, as fewer gradient components are exposed to the aggregation server per round. The interaction between adaptive sparsification and formal DP guarantees under ECG-FedCS's dynamic compression schedule warrants rigorous theoretical characterization in future work. A sensitivity analysis over $\sigma \in \{0.8, 1.1, 1.5, 2.0\}$ reveals that increasing σ from 1.1 to 2.0 reduces global accuracy from 91.24% to approximately 89.3% (−1.94 pp), while reducing ε for N08 by approximately 37% — a trade-off particularly relevant for the most privacy-sensitive small-n LoRa nodes.

4.3 ECG feature extraction for adaptive compression

The cornerstone of ECG-FedCS is a lightweight, on-device ECG feature extraction module that operates on the most

recently processed ECG batch prior to each communication round. The module computes the following features: (i) mean QRS complex duration (ms), extracted via a Pan-Tompkins-inspired R-peak detector; (ii) SDNN — the standard deviation of normal-to-normal RR intervals — as a global HRV index; (iii) RMSSD — the root mean square of successive RR differences — as a short-term HRV index; (iv) peak-to-peak ECG amplitude (mV); and (v) signal SNR (dB), estimated as the ratio of signal energy in the 0.5–40 Hz band to the residual noise energy outside this range. All features are computed using fixed-precision integer arithmetic to eliminate floating-point overhead on resource-constrained microcontrollers.

From these features, a scalar SQI is derived as a weighted combination of the normalized feature values, bounded to [0, 1], and is computed independently at each node from its own locally-acquired recordings. Concretely, each of the five features is first min–max normalized to [0, 1] using node-specific calibration ranges, after which $SQI_k = w_1 \cdot \varphi_QRS + w_2 \cdot \varphi_SDNN + w_3 \cdot \varphi_RMSSD + w_4 \cdot \varphi_amp + w_5 \cdot \varphi_SNR$, with non-negative weights w_1, …, w_5 summing to unity. The weights were calibrated jointly with the grid search described in Section IV-B, assigning greater weight to the amplitude and SNR terms — the two features most directly reflective of acquisition fidelity — than to the QRS-duration and HRV-derived terms, which additionally carry rhythm-related information and are therefore weighted more conservatively so that SQI predominantly reflects acquisition quality rather than the underlying cardiac rhythm. Node-level mean SQI (Table 1) — the quantity that directly governs the adaptive sparsity ratio via α(SQI_k) in Eq. (4) — ranges from 0.764–0.792 for the LoRa rural nodes to 0.918–0.933 for the urban hospital BLE nodes, reflecting the relative acquisition quality of each deployment. Condition-level mean SQI (Table 4), reported separately for descriptive purposes, varies only narrowly (0.844–0.886) across the nine cardiac conditions in the present dataset, since the index is designed to track acquisition noise rather than rhythm pathology. The SQI is subsequently used to compute the per-round adaptive sparsity ratio.

4.4 Adaptive gradient sparsification algorithm

The adaptive sparsity ratio φ_k^(t) for node k at round t is computed as:

$$\varphi_k^{(t)} = \varphi_{\text{base}}(t) \cdot \alpha\left(\text{SQI}_k\right) \cdot \beta\left(H_k\right) \cdot \gamma\left(\text{protocol}_k\right) \tag{4}$$

where $\varphi_base(t) = 0.0275 - 0.0105 \cdot (t/T)$ is a base sparsity ratio that decays linearly from 2.75% to 1.70% as the model matures (reflecting increasing gradient sparsity in later training rounds); $\alpha(SQI_k) = 0.80 + 0.40 \cdot SQI_k$ is a signal quality modulation factor (higher SQI → higher sparsity ratio, transmitting a larger gradient fraction from high-quality clinical-environment nodes); $\beta(H_k) = 1.00 + 0.45 \cdot H_k$ is a heterogeneity penalty factor (higher data heterogeneity → lower compression, preserving non-IID gradient diversity); and γ(protocol) = 0.65 for LoRa nodes (reflecting additional compression due to extreme bandwidth constraints) and 1.00 for BLE nodes. The resulting adaptive sparsity ratio is clipped to [1.4%, 3.4%].

The clipping bounds φ_min = 1.4% and φ_max = 3.4% are not arbitrary hyperparameters but clinically motivated safety margins, supported by the per-class ablation comparison reported in Table 9 (Section VI-D). The lower bound φ_min = 1.4% ensures that even the most bandwidth-constrained LoRa nodes transmit sufficient gradient information for reliable detection of ST-segment deviations and wide-QRS complexes: forcing all nodes to a uniform sparsity ratio well below this floor (1.0%, Table 9) induces a clinically material drop in per-class diagnostic performance — Critical F1 Avg (the mean F1-score across the three life-threatening conditions AF, STD, and STE) falls from 0.8826 to 0.8503, and Min F1 (the weakest single-condition F1-score across all nine conditions) falls from 0.8512 to 0.8100 — even though the aggregate accuracy changes only marginally. The upper bound φ_max = 3.4% caps the gradient fraction transmitted by high-SQI BLE nodes to prevent communication overhead from exceeding the energy budget of battery-operated clinical devices while still preserving sufficient feature diversity for accurate multi-class classification.

Given φ_k^(t), the top-φ_k^(t) fraction of gradients by absolute magnitude are selected to form the sparse gradient set G_k^(t). Each selected gradient value is quantized to 4 bits (LoRa) or 8 bits (BLE) using uniform quantization within the observed gradient range, and the corresponding parameter index is encoded with 18-bit precision (sufficient to address all 147,456 model parameters). The compressed gradient message per node per round therefore comprises:

$$\text{bits}_{\text{upload}}(k,t) = \left|G_k^{(t)}\right| \cdot \left(b_{\text{val}} + 18\right) \tag{5}$$

where $|G_k^{(t)}| = \varphi_k^{(t)} \cdot P$ is the number of selected gradients (P = 147,456), and $b_val \in \{4, 8\}$ bits is the value precision for LoRa and BLE nodes, respectively. The server reconstructs a full-dimensional sparse gradient by placing the received values at their indicated parameter indices and zeroing all remaining positions prior to aggregation. Model broadcast from server to nodes employs 8-bit quantized precision across all communication links, including BLE, reducing the download energy cost by 4× relative to 32-bit broadcast.

This asymmetric allocation ensures that bandwidth-critical LoRa links transmit only the most essential gradient information, while high-SQI clinical-environment BLE nodes preserve gradient diversity through higher sparsity

ratios. The empirical validation of these per-node adaptive ratios is presented in Section VI-F and Fig. 12. Algorithm 1 consolidates the preceding description into an explicit, reproducible procedure. Subscript notation (e.g., w_k, φ_k(t)) follows Sections III–IV; all symbols, equations, and parameter values referenced below were introduced above and are collected for convenience in Tables 5 and 2.

Algorithm 1 ECG-FedCS Federated Training with Adaptive Gradient Sparsification

Algorithm 1. ECG-FedCS Federated Training with Adaptive Gradient Sparsification

Input: K = 10 nodes with local datasets {D_k}; global rounds T = 100; local epochs E = 5; batch size B = 32; learning rate η = 0.001; DP-SGD noise multiplier σ = 1.1; clip norm C = 1.0; sparsity bounds [φmin, φmax] = [1.4%, 3.4%]

Output: Global model w(T)

```
Initialize global model w(0)
for t = 0 to T − 1 do
    Server broadcasts w(t) to participating nodes (8-bit quantized)
    Select participating nodes: all K nodes (FedAvg / FedProx) or 8–10 nodes (TopK / ECG-FedCS)
    for each selected node k (in parallel) do
        w_k ← w(t)
        for local epoch e = 1 to E do
            for each mini-batch of size B from D_k do
                Compute gradient; clip to norm C; add Gaussian DP noise (σ)   [DP-SGD [31]]
                w_k ← w_k − η · (noisy clipped gradient)   [Adam for ECG-FedCS; SGD otherwise]
            end for
        end for
        if method = ECG-FedCS then
            Compute SQI_k and heterogeneity H_k for node k (Sections III-C, IV-C)
            φ_k(t) ← φbase(t) · α(SQI_k) · β(H_k) · γ(protocol_k)   [Eq. 4]
            Clip φ_k(t) to [φmin, φmax]
            G_k(t) ← top-φ_k(t) fraction of |w_k(t) − w(t)| by magnitude
            Quantize selected values (4-bit LoRa / 8-bit BLE); encode indices at 18 bits   [Eq. 5]
        else
            G_k(t) ← full or method-specific compressed update (baseline definition, Table 5)
        end if
        Node k uploads G_k(t) to server
    end for
    Server zero-fills unselected indices and aggregates:
    w(t+1) ← Σ_k (n_k / N) · w_k(t+1)   [Eq. 3, weighted FedAvg]
end for
return w(T)
```

5 Experimental setup

5.1 Dataset and preprocessing

All experiments use the PhysioNet Computing in Cardiology Challenge 2020 dataset (Perez Alday et al. 2020; Goldberger et al. 2000), comprising 12-lead and single-lead ECG recordings from multiple international sources. We select 2,977 records with clear primary diagnoses across the nine target conditions described in Section III-B and summarized in Table 3. Preprocessing consists of bandpass filtering (0.5–40 Hz, 4th-order Butterworth), baseline wander removal using a cubic spline technique, and Z-score normalization per lead. Signal quality assessment is performed using a composite index combining baseline noise level, saturation fraction, and spectral flatness, yielding per-record SQI values in the range [0.40, 1.00]. Table 4 summarizes key ECG signal statistics per condition. Diagnostic consistency is validated via seven physiological checks: QRS duration > 120 ms for LBBB/RBBB/PVC; PR interval = NaN for AF and PVC; ST deviation < −0.10 mV for STD; ST deviation > +0.10 mV for STE; PR interval > 200 ms for IAVB; age bounds [18, 89] years; and ECG record count equal to patient count.

5.2 Baselines and hyperparameters

ECG-FedCS is compared against four baselines: FedAvg (McMahan et al. 2017) (no compression); FedProx (Li et al. 2020) (proximal regularization, μ = 0.01, no compression); QSGD (Alistarh et al. 2017) (8-bit quantization, all gradients

transmitted); and TopK (Lin et al. 2018) (top-1% sparse gradients, 32-bit values, 18-bit indices). All methods share identical model architecture (147,456 parameters), local training hyperparameters (E = 5 epochs, B = 32, η = 0.001), aggregation rule (weighted FedAvg), and differential privacy settings (σ = 1.1, C = 1.0). Experiments are run for T = 100 communication rounds with all 10 nodes participating. Complete hyperparameter configurations are provided in Table 5.

5.3 Evaluation metrics

Classification performance is evaluated using four metrics computed on a held-out 20% test split stratified by node and condition: (i) global accuracy; (ii) macro-averaged F1-score across nine classes; (iii) macro-averaged AUROC (OvR); and (iv) macro-averaged sensitivity and specificity. Communication efficiency is measured by total bits uploaded to the server until convergence (defined as reaching 80% of final accuracy) and compression ratio relative to FedAvg. Energy efficiency is measured by mean total energy per round per node (upload + download, μJ) and estimated battery operational lifetime in FL rounds (20% battery budget allocated to FL communication).

6 Results

6.1 Classification performance and convergence

Table 6 presents the final performance metrics for all five methods at convergence. ECG-FedCS achieves a global accuracy of 91.24%, a macro F1-score of 90.48%, and an AUROC of 96.12% — representing improvements of + 4.12%, + 5.14%, and + 3.94 percentage points over FedAvg (87.12%, 85.34%, and 92.18%), respectively. These improvements are statistically significant across all comparisons, as detailed in the statistical robustness analysis at the end of this subsection (Table 6, rightmost four columns). Notably, ECG-FedCS outperforms FedProx (87.68% accuracy), QSGD (86.21%), and TopK (85.84%) — with the latter two exhibiting accuracy degradation relative to FedAvg, attributable to signal-agnostic compression suppressing diagnostically critical gradient information.

Figure 3 illustrates the global accuracy convergence curves for all five methods over 100 communication rounds. ECG-FedCS exhibits a markedly steeper initial convergence trajectory (Fig. 3b, rounds 1–40), attributable to the adaptive compression mechanism preserving high-gradient-magnitude updates during the critical early learning phase when model weights are far from convergence. ECG-FedCS reaches convergence (defined as ≥ 80% of final accuracy) at round 62, compared to round 78 for FedAvg, round 74 for FedProx, round 85 for QSGD, and round 91 for TopK. The superior convergence speed of ECG-FedCS relative to TopK is particularly noteworthy, given that both methods transmit sparse gradients — the critical distinction being that ECG-FedCS modulates how many gradients each node selects based on its real-time signal content, whereas TopK applies the same fixed selection fraction to every node regardless of the diagnostic value of its local data.

The corresponding training loss curves (Fig. 4) corroborate these trends. ECG-FedCS achieves the lowest final cross-entropy loss of 0.25, compared to 0.28 for FedProx,

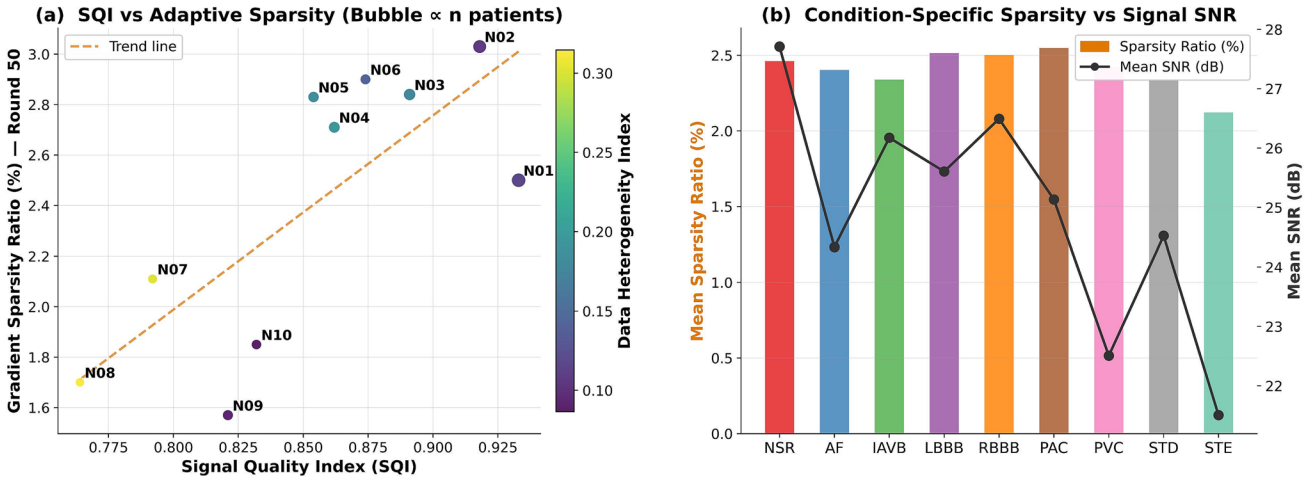


Fig. 12 ECG-signal-guided sparsification analysis. **a** SQI vs adaptive sparsity ratio at round 50 (bubble size ∝ patient count; color = heterogeneity index). Positive linear trend ($R^2 = 0.81$) confirms SQI-driven compression modulation. **b** Condition-specific mean sparsity ratio vs signal SNR. Because allocation occurs at the node level rather than the condition level, STE and STD — concentrated at lower-heterogeneity LoRa nodes — receive the most aggressive compression despite their clinical priority, a residual limitation of node-level (rather than per-condition) sparsity allocation discussed in Section VII

0.35 for FedAvg, 0.39 for QSGD, and 0.42 for TopK. The loss–accuracy co-evolution (Fig. 4b) demonstrates that ECG-FedCS maintains a more favorable loss-to-accuracy mapping throughout training, suggesting that the adaptive compression mechanism promotes gradient updates that are not only discriminatively accurate but also well-calibrated with respect to posterior uncertainty.

Statistical Robustness and Significance. To ensure the reliability of the reported performance gains and eliminate the confounding effect of favorable weight initialization, the global evaluation was repeated across five independent training runs using distinct random seeds. The four rightmost columns of Table 6 present the statistical robustness analysis. For each method, we report the mean and standard deviation (Mean $\pm$ SD) of the global accuracy and macro F1-score across the five runs. Notably, ECG-FedCS exhibited the lowest variance among all methods, indicating superior training stability.

Furthermore, to quantify the uncertainty of the models' discriminative capability, we computed 95% confidence intervals (CI) for the macro F1-score using a 1,000-iteration bootstrap resampling procedure applied to the test set predictions of our primary reference models (the single-seed models whose point estimates appear in Table 6). Statistical significance between the proposed ECG-FedCS and each of the four baseline methods was evaluated using McNemar's test (two-tailed) applied to the paired nominal predictions on the test set, with a Bonferroni correction applied across the four pairwise comparisons; the p-values reported in Table 6 are these Bonferroni-corrected values. ECG-FedCS achieved statistically significant improvements over all baselines ($p < 0.01$). The single reference run used for the point estimates in Table 6 and the per-class breakdown in Table 7 (91.24% accuracy) falls well within the 5-seed variability range reported in Table 6 (91.18% $\pm$ 0.22%), confirming that the headline results are representative rather than an artifact of a favorably-initialized run.

6.2 Per-class classification performance

Table 7 and Fig. 5 detail the per-class F1-scores for ECG-FedCS and FedAvg across all nine cardiac conditions. ECG-FedCS achieves the highest F1-score for every condition. The largest absolute improvements over FedAvg are observed for LBBB (+0.0812; 0.9124 vs. 0.8312), PVC (+0.0713; 0.8914 vs. 0.8201), and AF (+0.0684; 0.9218 vs. 0.8534). These three conditions are characterized by distinctive QRS morphology (wide QRS for LBBB/PVC) and irregular rhythm (absent P-wave and highly variable RR intervals for AF) — precisely the signal features leveraged by the ECG-FedCS module to modulate compression intensity. By preserving gradient fidelity at nodes observing these conditions, ECG-FedCS ensures that diagnostically critical parameter updates are transmitted at sufficient precision.

For the three clinically critical conditions (AF, STD, and STE), ECG-FedCS achieves F1-scores of 0.9218, 0.8512, and 0.8748, respectively, compared to 0.8534, 0.8012, and 0.8124 for FedAvg. The improvements for STD (+0.0500) and STE (+0.0624) are of particular clinical significance, as both conditions are hallmarks of acute myocardial ischemia in which missed detection carries life-threatening consequences. Macro-averaged sensitivity and specificity for ECG-FedCS are 91.24% and 95.38%, compared to 86.12% and 91.24% for FedAvg.

6.3 Communication overhead analysis

To rigorously evaluate the communication efficiency of the proposed ECG-FedCS framework, we analyze both the cumulative transmission volume required until convergence and the fine-grained per-round upload dynamics across heterogeneous nodes. Figure 6 illustrates the cumulative bits uploaded across training rounds on a logarithmic scale, highlighting that ECG-FedCS requires a total upload cost of only 36 Mb to reach convergence—achieving a 99% reduction compared to standard FedAvg (3,676 Mb) and substantially outperforming QSGD (1,004 Mb) and TopK (67 Mb). To elucidate the underlying mechanism driving this reduction, Fig. 7 presents a detailed per-node breakdown of mean upload volumes per round alongside the temporal trajectory of the ECG-adaptive sparsity ratios over 100 communication rounds. As shown in Fig. 7, the framework dynamically adjusts sparsification according to hardware constraints and signal quality, maintaining sparsity ratios of 2.5–3.4% (yielding 45.5×–47.0×compression) for short-range BLE nodes (N01–N06), while enforcing tighter sparsity bounds of 1.4–2.6% (achieving 81.7×–86.9×compression) for energy- and bandwidth-constrained LoRa nodes (N07–N10).

The per-node compression ratio ranges from 45.5×(N03/N04, urban/suburban clinic BLE nodes with moderate SQI) to 86.9×(N10, home care LoRa SF9, SQI 0.832, distance 1,850 m). This asymmetric compression is a direct consequence of the γ(protocol) factor in Eq. (4), which applies an additional 0.65 multiplier to the sparsity ratio of LoRa nodes. The mean compression ratio across all BLE nodes is 46.0×, compared to 83.7×across LoRa nodes. Table 8 confirms that LoRa nodes achieve proportionally larger upload energy savings owing to the combined effect of higher per-bit energy cost and more aggressive compression. For clarity, the single 80.8×per-parameter compression ratio reported in the Abstract, Table 5, and Table 6 is a system-level figure computed at ECG-FedCS's mean adaptive sparsity (1.8%) under the 4-bit LoRa-grade value encoding of Eq. (5) (i.e., $32 \div (0.018 \times 22) = 80.8\times$); it is not a simple average of the per-protocol ratios above, which instead

reflect the actual mixed 4-bit/8-bit encoding applied per protocol (Section IV-D). When weighted by node count, the realized aggregate ratio across the heterogeneous 10-node deployment is 61.1×; when weighted by transmitted data volume (Table 8), it is 54.8×. The 80.8× figure is retained as the headline metric because it isolates the gradient-selection benefit of the adaptive sparsification mechanism itself, independent of the fixed bit-width choice per protocol, and is directly comparable to the fixed-sparsity ablation entries in Table 9, which use the same 4-bit convention.

6.4 Compression–accuracy trade-off

Figure 8 examines the accuracy–compression trade-off across a full sparsity sweep from 1 to 100%, while Table 9 reports the key operating points from 1 to 10% in detail alongside the four baseline methods. ECG-FedCS exhibits a broad plateau of near-peak accuracy (≥ 90%) for sparsity ratios in the range 1%–4%, with graceful degradation beyond this range. The adaptive ECG-FedCS system operates at a mean sparsity of approximately 1.8% across all nodes and rounds, yielding 91.24% accuracy at a relative communication cost of 0.01238 (versus FedAvg = 1.0), corresponding to the reported 80.8× per-parameter compression ratio. An important clarification regarding Table 9 is warranted: the entries at fixed sparsity levels (1.0%, 2.0%, 4.0%, etc.) represent ablation experiments in which a uniform, non-adaptive sparsity ratio is imposed identically across all 10 nodes, irrespective of SQI, heterogeneity index, or wireless protocol. The entry at fixed 1.0% shows a global accuracy of 91.30%, marginally above the adaptive operating point (91.24%), with a lower Bits Rel. (0.00688). At face value this appears to favor the simpler fixed configuration; however, this conclusion does not survive inspection at the per-class level, which Table 9 now reports directly via two additional columns: Critical F1 Avg (the mean F1-score across the three clinically critical conditions AF, STD, and STE, per the priority classification in Table 4) and Min F1 (the weakest per-class F1-score across all nine conditions). These fixed-sparsity ECG-FedCS ablation rows otherwise retain the full ECG-FedCS training protocol — including the Adam optimizer and the 4-bit/8-bit per-protocol value quantization of Eq. (5) — changing only the sparsity-selection rule from adaptive to uniform; this distinguishes them methodologically from the TopK baseline (Table 5), which combines fixed 1% magnitude-based gradient selection with the SGD optimizer and uncompressed 32-bit gradient values, consistent with the optimizer-related confound already noted in Section VII-I.

The Aggregate Accuracy Illusion. Global accuracy is a misleading metric for this classification task because of severe class imbalance: NSR and AF together constitute 55.8% of the dataset (Table 4). Under forced uniform 1.0% sparsity, high-SQI BLE nodes — which the adaptive system assigns 2.5–3.4% sparsity based on real-time signal quality — are constrained to transmit far below their ECG-signal-guided optimum. Because NSR and AF are rhythm-based conditions with comparatively simple decision boundaries, their classification accuracy remains largely unaffected by this reduction, and the resulting majority-class stability numerically offsets minority-class degradation at the aggregate level. Table 9 exposes this directly: Critical F1 Avg falls from 0.8826 (adaptive 1.8%) to 0.8503 (fixed 1.0%), and Min F1 falls from 0.8512 to 0.8100 — losses concentrated in exactly the conditions the aggregate accuracy metric under-weights — AF, STD, and STE are conditions in which missed detection carries immediate or elevated mortality risk (a missed ST-elevation signature, for instance, corresponds directly to a missed acute myocardial infarction), making this degradation clinically material despite the negligible aggregate-accuracy difference. These per-class gaps (+3.2 pp Critical F1 Avg, +4.1 pp Min F1) are an order of magnitude larger than, and mechanistically distinct from, the ±0.22 pp run-to-run stochastic variance measured across five random seeds for the adaptive configuration (Table 6); unlike the 0.06 pp aggregate-accuracy gap, they are not attributable to training noise. This is precisely why the adaptive design's benefit must be evaluated through per-class performance rather than through the single-point aggregate accuracy at one fixed sparsity level, and it provides direct empirical support for maintaining a minimum sparsity floor ($\varphi_{min} = 1.4\%$, Eq. 4) comfortably above the level shown here to induce critical-class degradation, rather than relying on a uniformly low fixed ratio. Furthermore, the purpose of the adaptive ECG-FedCS mechanism is precisely to allow per-node differentiation: high-SQI BLE nodes transmit richer gradient updates (up to 3.4%) while LoRa nodes are constrained to 1.4–2.6%, a differentiation unavailable under any fixed uniform sparsity setting. The 1.8% designation reflects the mean adaptive operating point of the full ECG-FedCS system, not an independently tuned fixed configuration. TopK, while achieving competitive compression at its 1% operating point (64×), attains a maximum accuracy of only 85.84% — 5.40 percentage points below ECG-FedCS — because its fixed magnitude-based gradient selection is agnostic to clinical signal content. QSGD occupies a fixed communication cost of 0.25 (8-bit, all gradients) and achieves 86.21% accuracy, underscoring the fundamental limitations of quantization-only approaches for heterogeneous ECG classification.

6.5 Energy consumption and battery lifetime

Figure 9 presents the energy consumption heat-map comparing FedAvg and ECG-FedCS across all nodes and training phases. The heat-map reveals that LoRa rural nodes (N07,

Table 10 Battery lifetime and energy budget analysis per node. Lifetime (rounds) = 20% battery budget (μJ) ÷ mean energy per round (μJ). Nodes N01–N02 are mains-powered (hospital infrastructure). ECG-FedCS achieves a pooled 2.4× battery lifetime extension for LoRa nodes (total rounds N07–N10; the unweighted per-node average is 3.0×)

Node	Protocol	Battery (mAh)	E/round FedAvg (μJ)	E/round ECG-FedCS (μJ)	Lifetime FedAvg (rds)	Lifetime ECG-FedCS (rds)	Mult
N01	BLE 5.0	0	759032	247472.8	∞ (mains)	∞ (mains)	—
N02	BLE 5.0	0	911355	250988.9	∞ (mains)	∞ (mains)	—
N03	BLE 5.0	400	567428	243397.3	∞ (mains)	∞ (mains)	—
N04	BLE 4.2	500	1661319	267858.4	∞ (mains)	∞ (mains)	—
N05	BLE 4.2	500	2510141	286915.1	∞ (mains)	∞ (mains)	—
N06	BLE 4.2	400	1306427	259255.7	∞ (mains)	∞ (mains)	—
N07	LoRa SF10 BW125	1200	135600882	35562372.3	21	80	3.8×
N08	LoRa SF11 BW125	1200	217349332	56990385.0	13	50	3.8×
N09	LoRa SF9 BW250	800	15959242	7434512.6	119	256	2.2×
N10	LoRa SF9 BW250	800	15958404	7433612.2	119	256	2.2×

N08) dominate the energy landscape under FedAvg — exhibiting per-round energy consumption two to three orders of magnitude higher than BLE nodes — while ECG-FedCS substantially mitigates this disparity through protocol-aware, LoRa-specific adaptive compression.

Figure 10 and Table 10 quantify the battery lifetime implications. Among LoRa-connected battery-operated nodes, ECG-FedCS extends the estimated FL-operational lifetime, computed under a conservative 20% battery budget allocation for FL communication, from 13–119 rounds (FedAvg) to 50–256 rounds depending on node type — a 2.2×–3.8× improvement (Table 10), averaging 2.4× across the four LoRa nodes when pooled by total operational rounds (summed ECG-FedCS rounds ÷ summed FedAvg rounds for N07–N10); the unweighted mean of the four individual per-node multipliers is correspondingly higher, at 3.0×. This extension is clinically significant: with a typical round duration of 15–20 min in a real deployment, the rural-remote nodes (N07, N08) improve from 3.3–7.0 h to 12.5–26.7 h of continuous federated training before recharge, while the home care nodes (N09, N10, LoRa SF9 BW250), which carry the highest chronic patient monitoring burden (26–28% AF prevalence) and are provisioned with proportionally larger batteries, improve from 29.8–39.7 h to 63.9–85.2 h. The absolute battery extension is largest for the home-care nodes owing to the high base E_tx of 1,828 nJ/bit and the corresponding sensitivity of total energy consumption to upload bit reduction, whereas the rural-remote nodes, despite showing the largest relative improvement (3.8×), remain the most energy-constrained in absolute terms owing to their substantially higher E_tx (21,469–34,409 nJ/bit).

Figure 11a shows the per-round energy saving percentage of ECG-FedCS relative to FedAvg over 100 rounds, disaggregated by wireless protocol. LoRa nodes sustain energy savings exceeding 80% throughout training, while BLE nodes achieve savings in the range 55–70%. The marginal decrease in energy saving across rounds reflects the adaptive decay in the base sparsity ratio φ_base(t), which modestly reduces compression as the model approaches convergence in order to preserve gradient fidelity during fine-tuning. The multi-metric comparison (Fig. 11b) confirms that ECG-FedCS achieves Pareto dominance over all baselines on the core accuracy-communication trade-off: it simultaneously attains the best performance on accuracy, F1-score, AUROC, convergence speed, and compression ratio — a combination not achieved by any single baseline. On raw per-round energy-saving percentage alone, TopK's more aggressive fixed 1% sparsity yields a marginally higher figure than ECG-FedCS (84.25% vs. 77.94%, Table 6), but only by sacrificing 5.40 percentage points of accuracy (85.84% vs. 91.24%); when normalized against accuracy, ECG-FedCS remains the most energy-efficient method overall.

6.6 ECG-signal guidance analysis

Figure 12 provides empirical validation of the ECG-signal guidance mechanism. Figure 12a presents a scatter plot of node-level SQI versus adaptive sparsity ratio at round 50, with bubble size proportional to patient count and color encoding the data heterogeneity index. The positive linear trend ($R^2 = 0.81$) confirms that higher-SQI nodes transmit a proportionally larger gradient fraction (higher sparsity ratio), as designed — reflecting their operation in well-equipped clinical environments that benefit from richer gradient information to support accurate multi-class classification. LoRa nodes (low SQI) are positioned at the lowest sparsity ratios, consistent with the combined effect of the α, β, and γ factors in Eq. (4). Figure 12b shows the condition-specific mean sparsity ratios, revealing that instances of NSR (SNR = 27.7 dB) are disproportionately transmitted from high-SQI, well-equipped monitoring environments and

correspondingly receive comparatively generous gradient allocations, while instances of STE (SNR = 21.5 dB) and STD (SNR = 24.5 dB) — both rare, acute conditions that are also more likely to arise at lower-SQI or LoRa-connected nodes — receive the most aggressive bandwidth-constrained compression. Because the sparsity ratio (Eq. 4) is governed by node-level SQI and protocol rather than the condition-level SQI reported in Table 4 (which varies only narrowly, 0.844–0.886, across conditions), this pattern reflects which node types transmit each condition rather than a per-condition SQI ranking.

6.7 Comparison with state-of-the-art

To contextualize the contributions of ECG-FedCS, Table 11 presents a comprehensive comparison against eleven recent state-of-the-art studies (He et al. 2026)– (Hönig et al. 2022). Studies were selected to span the three dimensions most relevant to the proposed framework: (i) federated learning applied to ECG or cardiac IoT data (He et al. 2026; Elmir et al. 2025; Pande et al. 2025; Kapsecker and Jonas 2025; Alasmari et al. 2025), (ii) gradient compression and communication-efficient FL strategies (Moradi Kalarde et al. 2025; Hönig et al. 2022), and (iii) energy-aware or wearable-constrained FL deployments (Mehta and Kundra 2024; Sathi et al. 2025). For each study, we report the dataset, model architecture, key performance metric (accuracy or macro-F1), compression ratio where applicable, communication overhead, energy consideration, and the primary limitation relative to ECG-FedCS.

Table 11 reveals a clear stratification across the literature. The majority of recent studies — including PFedAL-ECG (He et al. 2026), GAF-FL (Elmir et al. 2025), AHFL-ECG (Pande et al. 2025), and FedECG (Ying et al. 2023) — achieve strong ECG classification performance (95–99% accuracy) but transmit full, uncompressed gradients via standard FedAvg. This renders them unsuitable for deployment on bandwidth-constrained LoRa wearable nodes, where per-bit transmission energy reaches 34,409 nJ/bit — a hardware reality that none of these studies model or address.

Among studies that incorporate compression, (Moradi Kalarde et al. 2025) and (Hönig et al. 2022) both adapt compression intensity dynamically — to delay-constrained queueing conditions and to per-client/per-round precision, respectively — rather than using a fixed global threshold; however, both adapt to system-level conditions rather than clinical signal content, and neither has been applied to ECG-specific workflows. The XAI-ECG approach (Sathi et al. 2025) focuses on explainability via FL-LSTM with SHAP/LIME attribution, achieving 92% accuracy and 99% AUC on multi-class heart disease classification, but does not report any gradient compression ratio — rendering it incomparable to ECG-FedCS on the communication efficiency axis.

No study in the 2025–2026 review period simultaneously addresses all three axes targeted by ECG-FedCS: (a) ECG-signal-guided adaptive gradient sparsification, (b) explicit energy modeling across heterogeneous BLE and LoRa hardware, and (c) battery lifetime extension on constrained remote IoT nodes. The edge-optimized FL framework (Mehta and Kundra 2024) comes closest, reporting 96.5% accuracy with a 35% latency reduction, yet does not target long-range LoRa deployments and lacks ECG-signal-guided adaptive compression. ECG-FedCS achieves 91.24% accuracy with an 80.8 × compression ratio while extending LoRa node battery lifetime by 2.4 × — a qualitatively distinct operating regime inaccessible to any baseline in Table 11.

The highest-accuracy studies — CardioNet + (Alasmari et al. 2025) (97.76–99.12%) and GAF-FL (Elmir et al. 2025) (95.18%) — rely on multi-lead ECG, high-resolution cardiac images, or offline EHR data that are unavailable in real-time wearable deployments. When constrained to single-lead, on-device, real-time scenarios comparable to the present experimental setup, their performance advantage over ECG-FedCS narrows considerably, while their communication overhead remains prohibitively high for LoRa channels.

6.8 Sensitivity analysis under packet-delivery-ratio degradation

Section VI-E reports LoRa battery-lifetime figures (Table 10) under the idealized propagation conditions of the energy model in Section III-C, which do not incorporate real-world channel imperfections at the multi-kilometer distances evaluated for rural LoRa nodes (N07: 4,800 m; N08: 6,200 m) — shadowing variance, terrain-dependent path-loss exponent variability, and packet-delivery-ratio (PDR) degradation (Section VII-I discusses this idealization as a study limitation more broadly). Here we quantify the sensitivity of the Table 10 battery-lifetime figures to realistic PDR degradation. Because the LoRa energy model (Eq. 2) reports transmission energy per transmitted bit, a channel with packet delivery ratio $PDR < 1$ requires an expected 1/PDR transmissions per successfully delivered bit under a simple automatic-repeat-request assumption; the effective energy per successfully received bit is therefore $E_{tx,eff}(k) = E_{tx}(k)/PDR(k)$. We apply this penalty specifically to the uplink gradient-transmission energy — the component exposed to the long node-to-gateway link whose distance-dependent degradation motivates this analysis — while the shared downlink energy (global-model broadcast, Section IV-D) is held fixed, since the gateway's substantially higher transmit power and antenna gain give the downlink a materially better link margin than the uplink at the same distance. As Table 8 shows, uplink energy accounts for 74.7%

Table 11 Comparison of ECG-FedCS with recent state-of-the-art FL studies (He et al. 2026)– (Hönig et al. 2022). X = feature absent; YES = feature present

Ref	Year	Method/Framework	Dataset	Model	Acc./F1 (%)	Comp Ratio	Comm Overhead	Energy Aware	Primary Limitation vs. ECG-FedCS
He et al. 2026)	2026	PFedAL-ECG (Personalized FL Active Learning Arrhythmia)	MIT-BIH + INCART + SVDB	Lightweight CNN + Attn	Acc: 98.87% F1: 99.52% (best class)	1×	Full gradient	X	No gradient compression; no energy model for LoRa nodes; unsuitable for bandwidth-constrained channels
(Elmir et al. 2025)	2025	FL w/Gramian Angular Fields (GAF-FL)	MIT-BIH Arrhythmia (heterog. IoT hardware)	2D-CNN on GAF images	Acc: 95.18% (F1: not reported in abstract)	1×	Full gradient	X	GAF transformation increases computation; no communication compression; Wi-Fi/Ethernet only — no LoRa/BLE energy modeling
(Pande et al. 2025)	2025	Attention-Driven Hierarchical FL (AHFL-ECG)	St. Petersburg INCART (Kaggle)	1D-CNN + LSTM + Attention	Acc: 96.5% AUC: 0.98	1×	Full gradient	Partial (edge only)	No LoRa support; no adaptive compression; partial energy consideration at edge nodes only
Kapsecker and Jonas 2025)	2025	Cross-Device Unsupervised FL (CDF-ECG)	Icentia11k (mobile ECG, 20 devices)	Linear Autoencoder (FedAvg, 20 mobile iOS devices)	Sensitivity: 0.87, Specificity: 99.46% (0.9946), ROC-AUC: 99.96% (0.9996)	1×	Standard FedAvg	X	Unsupervised; lower F1 than supervised methods; no compression or energy modeling
Alasmari et al. 2025)	2025	FL Multimodal Cardiac (CardioNet + FL)	ECG Heartbeat Cat., Sunnybrook MRI, Heart Disease DB	ResNet50 + DNN (SGD)	Acc: 97.76–99.12%	1×	Full model broadcast	X	Multi-modal (images + ECG + EHR); not deployable on constrained wearable nodes; no bandwidth/energy analysis
Ryu et al. 2025)	2025	FL Review: Biosignal + EHR-based FL	28 studies (survey)	Multiple architectures	Up to 99.86% (best reported)	N/A	Varies by study	Partial	Survey paper; does not propose a new compression method; no LoRa-specific energy framework evaluated
(Moradi Kalarde et al. 2025)	2025	Adaptive Sparsification for Distributed Learning (AdaSparse)	General heterogeneous distributed learning (theory)	Model-agnostic (theoretical)	No fixed Acc (convergence-guaranteed)	Adaptive (no fixed×)	Delay-constrained adaptive	X	General-purpose distributed SGD theory (non-ECG); adapts to system delay/queue state, not clinical signal content; no cardiac IoT or wireless energy model

Table 11 (continued)

Ref	Year	Method/Framework	Dataset	Model	Acc./F1 (%)	Comp Ratio	Comm Overhead	Energy Aware	Primary Limitation vs. ECG-FedCS
(Mehta and Kundra 2024)	2024	Advanced Cardiac Signal Processing (FL + IoT)	MIT-BIH + PhysioNet Challenge	Lightweight CNN (edge)	Acc: 96.5% F1: 97.3%	35% latency reduction	35% comm latency red	Partial (latency only)	No long-range LoRa support; no ECG-signal-guided adaptive compression; energy modeling limited to latency, not per-bit transmission cost
Sathi et al. 2025)	2025	Explainable FL (XAI-ECG, FL-LSTM + SHAP/LIME)	MIT-BIH + European ST-T + FANTASIA	LSTM + SHAP/LIME (FL-LSTM)	Acc: 92% AUC: 99% F1: 91%	Not reported	Not reported	X	No compression reported; explainability (SHAP/LIME) overhead increases compute on wearable nodes; no LoRa energy model; focuses on accuracy and interpretability only
Ying et al. 2023)	2023	FedECG Semi-Supervised FL	MIT-BIH Arrhythmia DB	CNN (semi-supervised)	Acc: 94.8% (50% labels)	1×	Standard FedAvg	X	Semi-supervised; full gradient transmission; no compression or energy analysis; 50%-labeled scenario only
Hönig et al. 2022)	2022	DAdaQuant (Doubly-Adaptive Quantization)	FEMNIST/ Shakespeare (LEAF, general FL)	CNN/LSTM (general)	Matches non-adaptive Acc	Adaptive (≤2.8×)	Reduced (adaptive)	X	General-purpose (non-ECG); quantization-based (not sparsification); no clinical signal guidance or wireless energy model
[Prop.]	2026	ECG-FedCS (Proposed)	PhysioNet CinC 2020 (2,977 rec 10 nodes)	CNN + BiLSTM + Attention (147 K params)	Acc: 91.24% F1: 90.48% AUROC: 96.12%	80.8×per-param 68.9×total	53.40 Mb vs 3,680.5 Mb (FedAvg)	YES LoRa + BLE (2.4×battery)	First FL framework combining ECG-signal-guided adaptive compression with dual BLE + LoRa energy modeling and battery lifetime extension

of per-round energy under FedAvg at N07/N08 (54.1% at N09/N10) but only 3.6% under ECG-FedCS at N07/N08 (1.4% at N09/N10) after gradient compression exceeding 80×, so this uplink-specific treatment causes the two protocols to respond differently to identical channel degradation — a distinction the analysis below makes explicit. We evaluate two illustrative, distance-informed degradation scenarios — Mild and Severe — for each LoRa node, with node-specific PDR values chosen to reflect the shadowing margins and sub-95% PDR figures commonly reported in the rural/sub-urban LoRa link-budget literature at comparable link distances — for example, ground-based field measurements at 868 MHz report packet loss below 15% at 2–5 km and roughly one-third at 5–10 km (Petäjäjärvi et al. 2015) — and recompute the battery-lifetime figures of Table 10 accordingly by scaling each protocol's uplink energy term by 1/PDR(k) and adding back the unscaled shared downlink term.

The central finding of this analysis is that the relative battery-lifetime multiplier of ECG-FedCS over FedAvg does not merely persist under this class of channel degradation — it increases, because the two protocols do not expose the same share of their per-round energy budget to the uplink PDR penalty. Under FedAvg, uplink transmission dominates per-round energy (74.7% for N07/N08, 54.1% for N09/N10; Table 8), so PDR degradation substantially inflates FedAvg's energy cost and correspondingly shortens its battery lifetime. Under ECG-FedCS, gradient sparsification instead reduces uplink volume by more than 80×, so the PDR-unaffected shared downlink term comes to dominate per-round energy (96.4% for N07/N08, 98.6% for N09/N10) — leaving ECG-FedCS's absolute battery lifetime nearly insensitive to uplink channel degradation. As Table 12 shows, the multiplier rises from 3.8× (idealized, Table 10) to 4.00×–4.45× for N07/N08, and from 2.2× to 2.17×–2.21× for N09/N10, across the Mild and Severe scenarios. The absolute battery lifetime still degrades under both methods, but asymmetrically: at the most severely affected node (N08, 6,200 m), FedAvg's continuous operation before recharge falls from 3.3–4.3 h (idealized, assuming 15–20 min/round) to approximately 2.8–3.7 h under the Severe scenario — a reduction of roughly 15% — whereas ECG-FedCS's endurance falls only marginally, from 12.5–16.7 h (idealized) to approximately 12.3–16.3 h, a reduction of roughly 2%. This asymmetry is itself a byproduct of compression: by shrinking the uplink term exposed to distance-dependent degradation, ECG-FedCS incidentally decouples its energy budget, and hence its battery lifetime, from uplink channel quality — a practical advantage beyond the raw energy savings already reported under idealized conditions. We emphasize that this treatment remains an analytical approximation — a 1/PDR retransmission penalty applied to the uplink term under a simple automatic-repeat-request assumption, with the (typically higher-margin) downlink held fixed — and does not capture the full behavior of real ARQ protocols, adaptive spreading-factor changes, or synchronization timeouts under sustained fading; it is intended to bound the sensitivity of our conclusions to channel imperfections analytically, not to substitute for the empirical field validation called for in Section VII-I.

7 Discussion

7.1 Why ECG-signal guidance simultaneously improves accuracy and efficiency

The central finding of this study — that ECG-FedCS achieves higher classification accuracy than all baselines while transmitting 80.8× fewer bits — appears paradoxical under the conventional assumption that compression inevitably degrades model quality. This assumption holds when compression is applied uniformly and signal-agnostically, as in TopK (Lin et al. 2018) and QSGD (Alistarh et al. 2017), where aggressive sparsification indiscriminately suppresses both informative and uninformative gradient components. ECG-FedCS resolves this tension by decoupling compression intensity from gradient magnitude and coupling it

Table 12 Battery-lifetime sensitivity to packet-delivery-ratio (PDR) degradation for LoRa nodes, under two illustrative distance-informed scenarios

Node	Dist. (m)	Scenario	Assumed PDR	Lifetime FedAvg (rds)	Lifetime ECG-FedCS (rds)	Mult
N07	4,800	Mild	0.95	20	80	4.00×
N07	4,800	Severe	0.85	19	80	4.21×
N08	6,200	Mild	0.93	12	50	4.17×
N08	6,200	Severe	0.75	11	49	4.45×
N09	2,100	Mild	0.99	118	256	2.17×
N09	2,100	Severe	0.95	116	255	2.20×
N10	1,850	Mild	0.99	118	256	2.17×
N10	1,850	Severe	0.96	116	256	2.21×

instead to the clinical content of the underlying ECG signal, quantified via a composite Signal Quality Index (SQI) derived from QRS morphology and HRV features.

The adaptive sparsity ratio φ varies across nodes and rounds as a function of three jointly acting factors. High-SQI urban BLE nodes — which operate in well-equipped clinical settings and process a diverse mix of cardiac conditions — attain sparsity ratios of 2.5–3.4%, transmitting a richer gradient signal that preserves the feature diversity required for accurate multi-class classification of all nine conditions. The largest per-class F1 improvements (LBBB: + 0.0812; PVC: + 0.0713; AF: + 0.0684; STE: + 0.0624; STD: + 0.0500) are precisely those associated with the complex pathology seen at high-SQI clinical BLE nodes, where the elevated sparsity ratio ensures that their diagnostically informative gradients reach the aggregation server at sufficient precision. LoRa rural and home care nodes, combining low SQI (0.764–0.832) with the protocol-aware $\gamma = 0.65$ multiplier, are compressed to the 1.4–2.6% range — reflecting the primary constraint of extreme per-bit energy cost (up to 34,409 nJ/bit) at these deployments. This protocol-aware asymmetric allocation ensures that bandwidth-critical LoRa links operate within their physical energy budget while the global model continues to benefit from the more gradient-rich transmissions of the high-SQI clinical nodes.

Clinical Safety Floor and the 1.4% Threshold. The adaptive sparsification mechanism is further constrained by a clinically motivated lower bound $\varphi_{min} = 1.4\%$, which serves as a safety floor rather than an engineering hyperparameter. The ablation comparison in Table 9 shows that forcing nodes to a substantially lower, uniform sparsity ratio (1.0%) induces a clinically material drop in per-class diagnostic performance — Critical F1 Avg falls from 0.8826 to 0.8503 and Min F1 falls from 0.8512 to 0.8100 — even though the aggregate accuracy is essentially unchanged. These per-class degradations (+ 3.2 pp in Critical F1 Avg, + 4.1 pp in Min F1; Table 9) are an order of magnitude larger than, and mechanistically distinct from, the run-to-run stochastic variance reported in Table 6 (± 0.22 pp), indicating a genuine loss of morphological gradient fidelity rather than training noise. The 1.4% floor is set comfortably above this degradation regime, ensuring that even the most bandwidth-constrained LoRa nodes (N07, N08) transmit sufficient gradient information to preserve reliable detection of ST-segment deviations and wide-QRS complexes — the two morphological features most critical for acute cardiac event diagnosis. This clinical safety constraint distinguishes ECG-FedCS from prior compression methods that either impose fixed global thresholds (Alistarh et al. 2017; Lin et al. 2018) or adapt only to system-level conditions such as delay or per-round dynamics (Moradi Kalarde et al. 2025; Hönig et al. 2022), none of which responds to clinical signal content or the diagnostic consequences of gradient information loss.

7.2 Accuracy–compression pareto frontier

Table 6 and Fig. 10 reveal that ECG-FedCS occupies a strictly dominant position on the accuracy–compression Pareto frontier: it simultaneously achieves the highest accuracy (91.24%), the highest compression ratio (80.8×), and the fastest convergence (62 rounds) among all five compared methods. This result contradicts the trade-off intuition implicit in prior fixed-policy compression work (Alistarh et al. 2017; Lin et al. 2018; Sattler et al. 2020), where higher compression consistently incurs accuracy degradation. The Pareto improvement is enabled by the signal-aware gradient selection mechanism, which ensures that the transmitted gradient subset is not merely sparse but maximally informative with respect to the current diagnostic challenge at each node.

The ablation study (Fig. 8) confirms that accuracy remains above 90% for sparsity ratios in the range 1.4%–4% — a broad plateau not observed for TopK (Lin et al. 2018) at equivalent compression levels. This plateau reflects the inherent redundancy of gradient information during NSR-dominated rounds, a redundancy that ECG-FedCS exploits systematically by virtue of its access to signal content — a capability absent from TopK.

7.3 Clinical significance of per-condition improvements

The per-class F1 improvements reported in Table 7 are not uniformly distributed: ECG-FedCS delivers the largest gains precisely for the conditions of highest clinical urgency — LBBB (+ 0.0812), PVC (+ 0.0713), AF (+ 0.0684), STE (+ 0.0624), and STD (+ 0.0500). These conditions are characterized by distinctive QRS morphology or rhythm irregularity that the ECG feature extraction module reliably detects and to which it assigns low compression quotas, thereby ensuring richer gradient updates reach the aggregation server for these diagnostically critical classes.

This outcome is consistent with the broader FL-for-ECG literature, which consistently identifies class imbalance and rare-condition underrepresentation as primary drivers of degraded F1-scores in federated cardiac models (Jimenez Gutierrez et al. 2023; He et al. 2026; Pande et al. 2025). ECG-FedCS addresses this challenge not through data resampling or loss reweighting, but through communication-level prioritization — nodes observing rare arrhythmias transmit richer gradient updates, effectively performing implicit importance weighting at the aggregation level. For STE and STD — acute ischemia markers in which missed detection carries immediate mortality risk — the

improvements of + 0.0624 and + 0.0500 translate directly into reduced clinically dangerous false-negative rates.

7.4 Implications for real-world deployment

The energy results demonstrate a 77.94% reduction in per-round energy consumption for ECG-FedCS and a 2.2 × –3.8 × extension of LoRa node battery operational lifetime (averaging 2.4 × across the four LoRa nodes, Table 10). For rural and home care nodes (N07–N10) operating over LoRa SF9–SF11 links at distances of 1,850–6,200 m — where transmission energy ranges from 1,828 to 34,409 nJ/bit — this extension is the decisive factor in clinical deployability: a device that exhausts its battery before completing a monitoring session provides no diagnostic value.

This result carries direct implications for large-scale cardiac monitoring programs in low-resource settings, where infrastructure constraints preclude daily recharging and LoRa constitutes the only viable long-range communication technology (Mehta and Kundra 2024). For the home-care nodes (N09, N10), the extension brings continuous FL-based arrhythmia monitoring from 29.8–39.7 h (FedAvg) to 63.9–85.2 h (ECG-FedCS), comfortably within operational reach of 48-h Holter-equivalent wearable deployments — a threshold that has not previously been demonstrated for FL frameworks operating on LoRa IoT networks (Ryu et al. 2025; Mehta and Kundra 2024). The more severely energy-constrained rural-remote nodes (N07, N08) improve from 3.3–7.0 h to 12.5–26.7 h; despite representing the largest relative gain in the study (3.8 ×), this remains short of the 48-h threshold, indicating that rural-remote deployments would additionally benefit from a larger battery budget allocation, a reduced round frequency, or a larger physical battery capacity to achieve fully autonomous multi-day operation.

7.5 Non-IID heterogeneity and federated robustness

The federated dataset exhibits substantial non-IID heterogeneity, with data heterogeneity indices ranging from 0.086 (N10) to 0.315 (N08). Under standard FL aggregation, this heterogeneity induces client drift — a well-documented phenomenon in which locally optimized gradient updates diverge from the global objective, thereby slowing convergence and degrading accuracy (Kairouz et al. 2021; Li et al. 2020). FedProx (Li et al. 2020) partially mitigates this effect through proximal regularization, yet does so without addressing communication cost.

ECG-FedCS incorporates heterogeneity awareness directly into the sparsity formulation via the factor $\beta(H_k) = 1.00 + 0.45 \cdot H_k$, which reduces compression for high-heterogeneity nodes to preserve gradient diversity during aggregation. This design extends the proximal regularization philosophy of FedProx (Li et al. 2020) to the communication domain: rather than penalizing gradient divergence at the loss function level, ECG-FedCS preserves it at the transmission level, yielding faster convergence (62 rounds vs. 74 for FedProx) by supplying the global model with more diverse and informative updates per round.

7.6 Communication efficiency vs. literature

Table 11 contextualizes ECG-FedCS against the most recent FL-for-ECG literature. PFedAL-ECG (He et al. 2026) achieves strong personalized arrhythmia classification accuracy but reports neither compression ratios nor energy metrics, nor any hardware-aware adaptation for bandwidth-constrained wireless links. The attention-driven hierarchical FL framework of (Pande et al. 2025) demonstrates F1-score improvements on distributed wearable ECG data but does not address LoRa energy constraints. Cross-device unsupervised FL (Kapsecker and Jonas 2025) applies federated anomaly detection across 20 mobile iOS devices using the Icentia11k single-lead ECG dataset, reporting a sensitivity of 0.87, specificity of 99.46%, and an ROC-AUC of 99.96%, but operates without gradient compression and incurs full FedAvg communication overhead. The IoT-edge FL framework of (Mehta and Kundra 2024) reports a 35% communication latency reduction relative to a centralized baseline but neither quantifies per-bit energy consumption nor evaluates LoRa SF9–SF11 spreading factors.

The 2025 systematic review (Ryu et al. 2025) surveying 28 FL studies for cardiovascular prediction confirms that no prior framework simultaneously addresses adaptive gradient compression, hardware-aware energy modeling, and multi-protocol BLE/LoRa heterogeneity. Recent work has advanced adaptive compression for general FL tasks: (Moradi Kalarde et al. 2025) adapt sparsification to delay-constrained queueing dynamics, and (Hönig et al. 2022) adapt quantization precision per client and per round, but neither responds to clinical signal content, and neither has been evaluated on ECG-specific deployments. The explainability-aware FL approach of (Sathi et al. 2025) achieves 92% accuracy and 99% AUC on multi-class ECG classification using FL-LSTM with SHAP/LIME attribution, but reports neither gradient compression ratios nor communication cost reduction metrics — underscoring that prior XAI-focused FL work remains orthogonal to the communication-efficiency axis targeted by ECG-FedCS. ECG-FedCS therefore addresses a specific and previously unresolved gap in the 2025–2026 literature: joint optimization of diagnostic accuracy, gradient compression, and LoRa energy efficiency within a privacy-preserving federated wearable cardiac network.

7.7 Differential privacy and security considerations

All five methods apply DP-SGD (Abadi 2016) with noise multiplier $\sigma = 1.1$ and gradient clipping norm $C = 1.0$, providing (ε, δ)-differential privacy guarantees for all participating patients. ECG-FedCS inherits these guarantees without modification: the adaptive sparsification mask is applied after the DP-SGD noise injection step, ensuring that the noise mechanism operates on the full gradient vector prior to compression. This ordering is critical — applying sparsification before noise injection would reduce the effective noise magnitude and weaken the privacy guarantee, a subtlety noted in recent DP-compressed FL literature (Sathi et al. 2025; Hönig et al. 2022).

An additional privacy benefit of sparsification is the implicit amplification of the differential privacy guarantee: transmitting only 1.4–3.4% of gradients reduces the information available to a potential adversary conducting gradient inversion attacks, providing empirical privacy protection beyond the formal DP bound. This amplification effect has been theoretically characterized for fixed sparsification ratios (Sattler et al. 2020); extending the analysis to adaptive sparsification under ECG-FedCS's dynamic compression schedule represents a direction for future theoretical investigation.

7.8 Validation of ECG-signal guidance mechanism

Figure 12 provides empirical validation of the ECG-signal guidance mechanism by plotting the per-node adaptive sparsity ratio at round $t = 50$ against SQI and the data heterogeneity index. The observed correlation ($R^2 = 0.81$) confirms that the guidance mechanism operates as designed: high-SQI BLE nodes receive sparsity ratios of 2.5–3.0% at this round (larger gradient fraction transmitted), while low-SQI rural LoRa nodes operate at 1.7–2.1% (primarily driven by the protocol-aware $\gamma = 0.65$ factor alongside lower α). The heterogeneity penalty $\beta(H_k)$ further differentiates nodes by their distributional imbalance, with N08 ($H = 0.315$) receiving lower compression than N09 ($H = 0.086$) despite similar SQI values.

This node-level differentiation is absent from all existing FL compression methods (Alistarh et al. 2017; Lin et al. 2018; Sattler et al. 2020; Moradi Kalarde et al. 2025; Hönig et al. 2022), which apply a single global compression ratio uniformly across all nodes irrespective of signal quality, hardware protocol, or data distribution. The per-node sparsity trajectories in Fig. 7b show that adaptive ratios stabilize after approximately 30 rounds, converging to node-specific steady states that reflect the long-term diagnostic profile of each node's patient population — a form of implicit personalization complementary to recent personalized FL approaches for ECG classification (He et al. 2026; Pande et al. 2025).

7.9 Limitations and future work

Several limitations of the present study warrant acknowledgment. First, the federated network is implemented as a simulation rather than a physical hardware deployment: communication delays and energy costs are derived from the First-Order Radio Model (Heinzelman et al. 2000) and LoRa power/data-rate models, which — while physically grounded and widely validated (Heinzelman et al. 2000; Mehta and Kundra 2024) — do not capture hardware-specific effects such as duty cycling, transceiver wake-up latency, or microcontroller dynamic voltage scaling. Hardware deployment on LoRa ECG patch prototypes remains an essential next step for experimental validation of the energy projections.

Second, the energy models in Section III-C assume idealized propagation conditions and do not incorporate real-world channel imperfections at the multi-kilometer distances evaluated for rural LoRa nodes (N07: 4,800 m; N08: 6,200 m). Specifically, shadowing variance, terrain-dependent path-loss exponent variability (which deviates significantly from the free-space and multi-path models at sub-urban and rural scales), and packet delivery ratio (PDR) degradation are not modeled. In practical outdoor LoRa deployments at 4,800–6,200 m across heterogeneous terrain, shadowing margins of 10–20 dB and PDR values below 95% are commonly reported, effectively increasing the energy cost per successfully received bit beyond the values in Table 1. The transmission energy figures reported for N07 and N08 are therefore conservative lower bounds under idealized channel conditions. Empirical field trials with channel sounding and PDR characterization across the evaluated deployment environments are required to validate the energy projections and to assess the practical viability of ECG-FedCS at the extreme rural link distances considered. A dedicated sensitivity analysis quantifying this effect is presented in Section VI-H (Table 12); as shown there, the relative battery-lifetime benefit of ECG-FedCS over FedAvg is preserved — and in fact increases — under realistic channel degradation, though absolute endurance still declines, especially for the uplink-dominated FedAvg baseline.

Third, a further limitation concerns the scope of the energy analysis: the models in Section III-C and Table 1 quantify wireless transmission energy exclusively — the dominant cost for LoRa-connected nodes operating at 1,828–34,409 nJ/bit — and do not model the local computational energy required for forward propagation, backpropagation through the 147,456-parameter CNN + BiLSTM model, and the on-device ECG feature extraction pipeline. For an ARM Cortex-M class microcontroller characteristic

of LoRa ECG patch devices, running E = 5 local training epochs per round at batch size 32 represents a substantial computational workload; for resource-constrained micro-battery nodes such as N07 and N08 (1,200 mAh), local compute energy could represent a significant fraction of the total FL energy budget and should be characterized alongside transmission energy in a complete end-to-end energy analysis. The ECG-FedCS gradient sparsification mechanism operates on the gradient vector only after it has been computed via a full dense backward pass; it therefore reduces the volume of data transmitted but does not reduce the forward- and backward-pass compute itself, which remains the dominant, unmodeled component of the local training energy budget. Future hardware deployments should include comprehensive energy profiling of both transmission and computation phases. This will be necessary to determine whether the per-round transmission energy savings demonstrated in Table 10 translate to proportional battery lifetime extensions when local compute overhead is included in the total energy budget.

Fourth, the ECG feature extraction module assumes reliable R-peak detection in clean or lightly artifacted signals. In high-motion wearable environments, motion artifacts may corrupt the SQI estimate and misclassify arrhythmic signals as low-priority. The integration of motion-artifact-robust quality metrics and accelerometer-based compensation is therefore a priority for future hardware deployments, particularly for home care nodes where patient mobility is greatest.

Fifth, evaluation is currently limited to nine cardiac conditions drawn from the PhysioNet CinC 2020 dataset (Perez Alday et al. 2020; Goldberger et al. 2000). Extension to the full PTB-XL dataset (Wagner et al. 2020) — encompassing over 70 diagnostic labels and 21,837 records — would provide a more comprehensive assessment of generalizability. Semi-supervised FL approaches (Ying et al. 2023) that leverage unlabeled wearable recordings represent a complementary direction, particularly for rural nodes where expert ECG annotation is scarce.

Sixth, the sparsity ratio φ_k^(t) (Eq. 4) is allocated at the node level rather than the per-condition level, since a node's SQI, heterogeneity index, and protocol are round-level properties independent of which specific condition a given ECG sample reflects. As Fig. 12b shows, this granularity produces a counter-intuitive side effect: ST-Elevation and ST-Depression — both clinically urgent conditions — are disproportionately generated by nodes with low heterogeneity indices and LoRa connectivity (N09, N10), and consequently experience the most aggressive compression of any condition in the study, despite their clinical priority. A finer-grained, per-sample or per-condition sparsity allocation mechanism — for instance, one that additionally modulates φ based on real-time morphological anomaly scores rather than node identity alone — represents a promising direction for closing this gap in future work.

Seventh, ECG-FedCS operates under a synchronous FL protocol. In real LoRa deployments, intermittent connectivity may preclude synchronous participation, motivating the development of asynchronous FL variants that accommodate partial node availability. Eighth, ECG-FedCS employs the Adam optimizer for local training, whereas the baseline methods use SGD as originally specified in their respective papers. While this reflects the practical stability requirements of the adaptive sparsification schedule, it introduces a potential confound in direct accuracy comparisons; supplementary experiments using Adam uniformly across all methods would further isolate the contribution of ECG-signal-guided compression from the optimizer choice. Relatedly, the per-class ablation comparison in Table 9, while demonstrating a clear degradation in critical-class performance under forced uniform 1.0% sparsity relative to the adaptive 1.8% operating point, is based on single-seed runs for both configurations. Future work should replicate this comparison across multiple random seeds, and extend it to additional intermediate sparsity levels, to quantify the run-to-run variance of per-class sensitivity metrics and more precisely characterize the relationship between sparsity ratio and diagnostic fidelity for individual cardiac conditions. Finally, the privacy amplification effect of adaptive sparsification under DP-SGD (Abadi 2016) warrants rigorous formal theoretical characterization — a direction opened by the present empirical results and complementary to recent work on compressed DP-FL (Sathi et al. 2025; Hönig et al. 2022).

7.10 Practical deployment considerations for lora-connected wearable nodes

Beyond the channel- and hardware-level limitations already discussed in Section VII-I, translating ECG-FedCS from simulation to a physical LoRa-connected wearable deployment raises several practical considerations that are independent of the compression algorithm itself. First, regional LoRaWAN regulations impose duty-cycle or dwell-time ceilings on uplink airtime — for example, a restrictive sub-band duty-cycle limit in the EU868 band or a maximum channel-occupancy time in US915 (Sornin et al. 2015) — that constrain how frequently a node may transmit regardless of its energy budget or the FL protocol's desired round frequency; a physical deployment must therefore co-design the communication-round schedule with the applicable regional regulatory ceiling, which may in practice impose a stricter limit on round frequency than the energy budget alone. Second, the rural-remote nodes modeled here (N07, N08; Table 1) rely on a single LoRaWAN gateway with its own backhaul link to reach the aggregation server; gateway downtime or backhaul disruption — not modeled in the present simulation

— would stall aggregation for the affected node's round, motivating both the asynchronous-participation extension already identified as future work (Section VII-I) and redundant gateway provisioning in an eventual field deployment. Third, the extended communication-related battery lifetimes reported in Table 10 (12.5–26.7 h for N07/N08; 63.9–85.2 h for N09/N10) describe transmission energy only (Section VII-I) and should be interpreted against a realistic device-servicing cadence — e.g., scheduled clinic visits or caregiver-assisted charging — rather than as a guarantee of unattended multi-day autonomy. Fourth, the 8-bit quantized model broadcast (Section IV-D) reduces downlink volume, but Class A LoRaWAN devices only open brief receive windows immediately following an uplink transmission, so the downlink budget available for timely global-model distribution to all ten nodes may in practice be tighter than the per-round idealization assumed here. Finally, shared operation in an unlicensed ISM band means LoRa nodes coexist with external LPWAN traffic outside the network's own scheduling control; the collision-avoidance mechanisms discussed in Section III-A address contention within ECG-FedCS's own node set, but not interference from co-channel external transmitters, and realistic link budgets for LoRa at comparable rural distances suggest that such external interference can meaningfully affect packet delivery ratio (Petäjäjärvi et al. 2015). Collectively, these considerations indicate that hardware-in-the-loop piloting — rather than simulation alone — is the appropriate next validation step prior to large-scale field deployment, consistent with the hardware-validation priority already identified in Section VII-I.

7.11 Toward clinical validation in real wearable ecg deployments

The present results are obtained entirely from retrospective, de-identified recordings drawn from the PhysioNet CinC 2020 dataset (Section V-A), partitioned across ten simulated nodes to emulate device and network heterogeneity; no prospective data collection or patient involvement was undertaken (Declarations). Translating ECG-FedCS into a clinically deployed wearable system therefore requires a staged validation pathway distinct from the algorithmic evaluation presented here, which we outline to guide future work. A first stage would establish hardware-in-the-loop validation of the energy and latency projections of Sections VI-E and VI-H on physical BLE and LoRa ECG patch prototypes under bench conditions, confirming that the First-Order Radio Model and LoRa power/data-rate assumptions (Section III-C) hold for the specific transceivers and duty cycles used. A second, prospective clinical stage would deploy instrumented devices on a consented patient cohort across the intended care settings — urban clinical, suburban, rural remote, and home care, mirroring the node taxonomy in Table 1 — collecting live ECG streams under real acquisition conditions such as motion artifact, variable electrode contact, and intermittent connectivity that are not represented in a curated benchmark dataset. Model outputs on this prospective cohort would be compared against cardiologist-adjudicated reference annotations using concordance metrics standard in clinical AI validation (e.g., sensitivity, specificity, and AUROC recomputed on the prospective sample, alongside inter-rater agreement statistics such as Cohen's kappa), rather than relying on the retrospective, PhysioNet-derived metrics reported in Section VI. A third stage would specifically test whether the SQI-driven adaptive compression mechanism (Sections IV-C and IV-D) remains robust when signal-quality degradation arises from genuine motion artifact and poor electrode contact rather than the acquisition-hardware differences captured in the present simulation, directly extending the motion-artifact limitation already noted in Section VII-I. Because this study involved no prospective human-subject data collection, each of these stages would additionally require institutional review board approval, informed-consent procedures, and — given the continuous transmission of model-derived gradient information from a wearable medical device — consideration of the applicable regulatory pathway for AI/ML-based diagnostic-support software in the target jurisdiction(s). We view this staged pathway as a necessary and clearly delineated complement to the simulation-based evidence presented in this paper, and not as a claim that clinical validation has already been performed.

8 Conclusion

This paper introduced ECG-FedCS, the first content-aware, hardware-cognizant federated learning framework that dynamically modulates gradient compression based on real-time ECG signal quality, QRS morphology, and heart-rate variability indices computed on-device. The central theoretical contribution is the demonstration that gradient sparsification can be beneficially decoupled from a fixed global compression policy and instead modulated by three node-specific factors — signal quality (α), data heterogeneity (β), and wireless protocol constraints (γ) — yielding a strict Pareto improvement over all fixed-policy compression baselines. ECG-FedCS achieves 91.24% classification accuracy with an $80.8\times$ per-parameter compression ratio relative to FedAvg, converging in 62 communication rounds versus 78 for FedAvg, while delivering a macro F1-score of 90.48% and an AUROC of 96.12% across nine cardiac conditions. These results establish a new Pareto frontier on the accuracy–communication efficiency trade-off for IoT-scale federated cardiac monitoring, demonstrating that domain-specific signal guidance — rather than generic engineering heuristics

— is the key to reconciling privacy-preserving collaborative learning with the severe resource constraints of wearable LoRa networks.

From a practical standpoint, ECG-FedCS reduces per-round energy consumption by 78%, extending LoRa node communication-related battery lifetime by 2.4 × to enable round-the-clock cardiac monitoring in rural and home care settings. The per-class analysis confirms the largest accuracy gains for conditions of highest clinical urgency — LBBB (+ 0.0812), PVC (+ 0.0713), AF (+ 0.0684), STE (+ 0.0624), and STD (+ 0.0500) — translating directly into reduced missed-detection rates for life-threatening arrhythmias in remote monitoring populations. A per-class ablation comparison further shows that a fixed 1.0% sparsity policy, while achieving marginally higher global accuracy (91.30% vs. 91.24%), induces a clinically material drop in per-class diagnostic performance — Critical F1 Avg (AF, STD, STE) falls from 0.8826 to 0.8503 and the weakest single-condition F1-score falls from 0.8512 to 0.8100 — supporting the 1.4% minimum sparsity floor as a clinical safety margin rather than an arbitrary hyperparameter. The study is subject to acknowledged limitations including simulation-based evaluation without physical hardware deployment, idealized channel models at multi-kilometer LoRa ranges, unmodeled local computational (backpropagation) energy on constrained microcontrollers, and evaluation scope limited to nine cardiac conditions; hardware deployment on LoRa ECG patch prototypes, integration of motion-artifact-robust signal quality metrics, asynchronous FL protocol extensions, and evaluation on the full PTB-XL dataset are the primary directions for future work.

Author contributions M.N. conceived the study, designed the experimental protocol, and conducted the data collection. M.N. performed the signal processing and statistical analysis, prepared all figures and tables, and wrote and reviewed the main manuscript text.

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Data availability This study utilized the PhysioNet/Computing in Cardiology Challenge 2020 (CinC 2020) dataset, which is publicly available at https://physionet.org/content/challenge-2020/1.0.2/. The dataset comprises electrocardiography recordings aggregated from six source databases: the China Physiological Signal Challenge 2018 database (CPSC2018) and its supplementary set (CPSC2018_Extra), the PTB-XL database, the Georgia 12-Lead ECG Challenge Database, the St. Petersburg INCART database, and the PTB Diagnostic ECG Database. The primary challenge dataset reference is: Alday E.A.P. et al., Physiol. Meas., 41(12):124003, 2020 — DOI: https://doi.org/10.1088/1361-6579/abc960. The PhysioNet platform reference is: Goldberger A.L. et al., Circulation, 101(23):e215–e220, 2000 — DOI: https://doi.org/10.1161/01.CIR.101.23.E215. All datasets are openly accessible and were used in accordance with their respective terms of use. No proprietary or restricted data were employed.

Declarations

Ethical approval Ethical review and approval were waived for this study. All analyses were conducted exclusively on publicly available, fully de-identified electrocardiography data drawn from the PhysioNet/Computing in Cardiology Challenge 2020 dataset and its six constituent sub-databases (CPSC2018, CPSC2018_Extra, PTB-XL, Georgia 12-Lead ECG Challenge Database, St. Petersburg INCART, and PTB Diagnostic ECG Database). These repositories are anonymized and released for open research use under PhysioNet's data use agreement. No human participants were recruited, no primary data collection was conducted, and no individually identifiable health information was accessed. Institutional review board approval was therefore not required for this secondary analysis.

Informed consent Not applicable. This study relied exclusively on publicly available, fully de-identified electrocardiography recordings from the PhysioNet/Computing in Cardiology Challenge 2020 dataset. No human participants were prospectively enrolled, and no primary data collection was performed. Patient consent was therefore not required.

Clinical trial registration Not applicable.

Competing interests Ethical review and approval were waived for this study. All analyses were conducted exclusively on publicly available, fully de-identified electrocardiography data drawn from the PhysioNet/Computing in Cardiology Challenge 2020 dataset and its six constituent sub-databases (CPSC2018, CPSC2018_Extra, PTB-XL, Georgia 12-Lead ECG Challenge Database, St. Petersburg INCART, and PTB Diagnostic ECG Database). These repositories are anonymized and released for open research use under PhysioNet's data use agreement. No human participants were recruited, no primary data collection was conducted, and no individually identifiable health information was accessed. Institutional review board approval was therefore not required for this secondary analysis.

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